Last fall degrees of elliptic curve summation systems: machine-checked bounds and exact values

Adam Buran
a.buran28@gmail.com

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Abstract

The last fall degree (LFD) of a polynomial system controls, together with the number of solutions, the degree at which linear algebra solves it. We study it for the summation-polynomial systems of elliptic-curve index calculus in three standard presentations: prime-field curves whose factor base is a subgroup of order 2^k, written through squaring chains as by Petit, Kosters and Messeng, and binary and Koblitz curves with linear coordinate spaces. We prove upper bounds and check them in Lean 4: with m summands the prime presentation has LFD $\leq mk+1$, so $2k+1$ for two summands, at every depth and for every prime, curve and target. A proof on paper improves the two-summand bound to $k+2$ for nonsingular curves. We also check lower bounds in Lean: LFD ≥ 4 at every depth for two-summand systems with unit ideal and $3t^2+a \neq 0$, and LFD ≥ 7 at every depth $k \geq 3$ for three summands when $5t^3+3at+6b \neq 0$. We then compute exact values with certificates. On a grid of 3005 systems with random and symmetric factor bases every value is certified. For two summands over prime fields and generic targets, the value is 4 for $k \leq 5$, 5 for $6 \leq k \leq 9$, 6 for $10 \leq k \leq 13$ and 7 at $k = 14$ on the instances where it was determined. This agrees with the degree-of-regularity table of Petit, Kosters and Messeng and extends it by three depths. The upper bound at $k = 14$ comes from a degree-lifting computation on three instances; it has a written proof and passed a control on a curve with solutions, but has no independent certificate. The values fit $\mathrm{LFD}(k) = \lfloor (k+14)/4 \rfloor$, which we state as a conjecture. On the binary and Koblitz grids computed, every value is at most 5 for dimensions up to 10; this is not a bound, since the inverse family of the companion manuscript, on a Koblitz curve, has value 6 at dimension 10. For two summands the measured values are thus far below the proved bounds, and we describe what a tight bound would need. With three to eight summands the picture is different: at every depth $k \leq m-1$ computed the value is exactly $mk+1$, so the bound is attained, with values up to 25. We make no claim about the cost of computing discrete logarithms.

1 Introduction

The last fall degree, introduced by Huang, Kosters and Yeo [10] and developed by Huang, Kosters, Yang and Yeo [1], is the degree-charged invariant that, together with the number of solutions, controls the linear-algebra solving of zero-dimensional Weil-descent and summation-polynomial systems [1, Prop. 2.8]: for an input list F it is the least d such that every ideal element h is derivable from F inside the iterated degree-$\max(d, \deg h)$ closure. Upper bounds on it bound the Macaulay degree at which linear algebra solves the system, and the invariant depends on the presentation, not only on the ideal. For the summation-polynomial systems of index calculus on elliptic curves, three presentation families recur: prime-field curves with a factor base given by a multiplicative subgroup of $\mathbb{F}_p^*$ of two-power order, written through squaring chains (Petit, Kosters and Messeng [3]); binary curves with an $\mathbb{F}_2$-linear coordinate space, descended to Boolean coordinates (Faugère, Perret, Petit and Renault [4]; Semaev [5]);

and Koblitz curves, the binary curves with $b = 1$ which are defined over $\mathbb{F}_2$, so that the Frobenius map is an endomorphism of the curve, and for which Frobenius-invariant coordinate spaces are the natural *symmetric* factor bases.

Results. We call a statement *kernel-checked* when its proof has been verified by the Lean 4 kernel (with Mathlib and no additional axioms), *written* when it has a proof on paper that is not formalized, and *exact* when it is the outcome of a computation in exact arithmetic that ends with a certificate.

Upper bounds. The chain equations of the prime presentation have pairwise coprime leading terms, so they form a Gröbner basis [9, Ch. 2, §9]. We give a kernel-checked proof of the consequence we need (Theorem 2) and deduce that every input list containing the chains has LFD $\leq \max(mk + 1, \delta)$ (Theorem 3). For the normalized two-summand presentation this is $2k + 1$ at every depth $k \geq 2$, for every prime $p \geq 5$, curve and target (Corollary 4). As a bound this is the standard dimension bound of a finite quotient; the contribution is its formal proof in this generality, which also covers the binary and Koblitz presentations (Section 4). A written proof improves the two-summand bound to $k+2$ for nonsingular curves (Theorem 9).

Lower bounds. The following are kernel-checked: LFD ≥ 4 at every depth for every two-summand prime chain system with unit ideal and $3t^2 + a \neq 0$ (Theorem 5); LFD ≥ 7 at every depth $k \geq 3$ for the direct three-summand system with $5t^3 + 3at + 6b \neq 0$ (Theorem 11); LFD ≥ 5 at every depth $k \geq 8$ for two explicit two-summand polynomial systems, one of them a subgroup system for $k \leq 13$ (Theorem 13), using an equivalence between the closure modulo the chains and the ordinary closure (Theorem 12); the persistence of an exclusion $1 \notin V_D$ at all larger depths (Theorem 15); and one exact value above 4, LFD $= 5$ at depth 6 for an explicit instance, with both directions checked (Theorem 14).

Exact values. Sections 5 and 6 report the exact LFD of 3005 distinct systems from all three families, with random and symmetric factor bases, each value certified. For two summands over prime fields, larger computations (Section 7) extend the range to depth 14. On the instances where the exact value was determined it is 4 for $k = 2, \ldots, 5$, 5 for $k = 6, \ldots, 9$, 6 for $k = 10, \ldots, 13$ and 7 at $k = 14$ (Table 6, Figure 1), apart from the exceptional systems described in Section 6. Through $k = 11$ these are the values of the averaged degree of regularity that Petit, Kosters and Messeng report for the same presentation [3, Table 4]; the values at $k = 12, 13, 14$ lie beyond their table. The upper bound LFD ≤ 7 at $k = 14$ comes from a degree-lifting computation on three instances. It has a written proof and passed a control on a curve with solutions, but it has no independent certificate, and we say so wherever the value is used. The values fit $\mathrm{LFD}(k) = \lfloor (k + 14)/4 \rfloor$, which we state as a conjecture. With three summands the value is 7 at $k = 3, 4, 5$ on every instance computed and at least 8 on one instance at $k = 6$. With $m = 3, \ldots, 8$ summands (the summation polynomials S_4 to S_9) and depth $k \leq m - 1$, the value is exactly $mk + 1$ on 409 of 412 instances, so the dimension bound is attained by summation systems (Table 7). For binary and Koblitz curves the two-summand value is at most 5 on the grids computed, for coordinate spaces of dimension up to 10; the structured inverse family, on a Koblitz curve, reaches $\lfloor k/2 \rfloor + 1$, which is 6 at $k = 10$.

Explanations. The step of the prime value at depth 6 is the point at which a quadratic closure dimension, measured at $k = 6, \ldots, 13$ and bounded above for $k \geq 8$ by the kernel-checked reduction theorem applied to a computed base, falls below the quartic dimension of the degree-4 part of the ideal. On binary curves the apparent difference between random and Frobenius-invariant factor bases follows the number of solutions. Section 7 states what a tight upper bound would need.

These are statements about specific polynomial encodings. They imply nothing about relation yield, the rank of the relations or the cost of computing a discrete logarithm.

2 Setting and conventions

For a field K, a polynomial ring $A = K[Z_1, \dots, Z_r]$ and a finite input set $F \subset A$, let $V_{F,D}$ be the least K-linear subspace of A containing every input of total degree at most D and closed under $h \mapsto gh$ whenever $\deg g + \deg h \le D$; the multiplication rule applies to newly derived rows. Define

$$\mathrm{LFD}(F) = \min\{d : \ h \in \langle F \rangle \Longrightarrow h \in V_{F,\max(d,\deg h)}\}.$$

This is the ordinary iterated closure of [1]; the convention permits zero for proper ideals and is not the degree at which a particular solver stops [2]. When we call a computed value *generic*, we mean that it was found on random instances (random curves and random rational targets of large order, not of order two, with $3t^2 + a \neq 0$); this is not a claim about a Zariski-open set of coefficients, which is open (item 1 of Section 7). The Lean development formalizes $V_{F,D}$ as an inductive predicate (`InClosure`) over `MvPolynomial`, and $\mathrm{LFD}(F) \le d$ as `HasLFDBound F d`.

Finite-quotient theorem. Let $G \subseteq F$ with $0 \notin G$, and say G is *reduced-ideal trivial* when every $p \in \langle G \rangle$ whose monomials are all standard for G (not divisible by the graded-lex leading monomial of any $g \in G$) is zero. If every standard exponent has degree at most s and every input has degree at most δ, then $\mathrm{LFD}(F) \le \max(s+1, \delta)$. The proof constructs unique normal forms from graded-lex division and shows that the degree-$\max(s+1,\delta)$ closure, intersected with normal forms, is stable under multiplication by variables, hence is an ideal of the finite quotient containing the images of all inputs. This is `FiniteQuotient.finite_quotient_lfd_of_reduced_zero`; the Boolean instance with $G = \{Z_i^2 - Z_i\}$ and $s = r$ is `GridQuotient.boolean_lfd_bound`.

Presentations. *Prime chains.* Let $p > 3$ be prime with $2^k \mid p - 1$ and let $H \subset \mathbb{F}_p^*$ be the subgroup of order $N = 2^k$. A summand coordinate $x_i \in H$ is written through chain variables $u_{i,0}, \dots, u_{i,k-1}$ with

$$u_{i,j}^2 = u_{i,j+1} \ (0 \le j < k-1), \qquad u_{i,k-1}^2 = 1, \qquad x_i = u_{i,0},$$

so that $u_{i,j} = x_i^{2^j}$ and a monomial x_i^e with $e < N$ is the squarefree chain monomial given by the binary digits of e. The chain equations are the mk polynomials $u_{i,j}^2 - u_{i,j+1}$ and $u_{i,k-1}^2 - 1$. The summation polynomial $S_{m+1}(x_1, \dots, x_m, t)$ of a target abscissa t is reduced by the chains (the *normalized* generator); for $m = 2$ it has chain degree at most 2. A *coset* factor base cH is written through the same chains with $x_i = c\,u_{i,0}$, which rescales the coefficients of the generator and nothing else.

Binary and Koblitz descents. Let $K = \mathbb{F}_{2^n}$ and $E : y^2 + xy = x^3 + ax^2 + b$, $b \neq 0$; Koblitz curves are the cases $a \in \{0, 1\}$, $b = 1$. For $\mathbb{F}_2$-linear coordinate spaces W_i with explicit bases, write $x_i = \sum_j \beta_{i,j} Z_{i,j}$ with Boolean coordinates $Z_{i,j}$, expand S_{m+1} in a field basis and reduce every coordinate modulo $Z^2 - Z$. The coordinate polynomials together with all Boolean equations form the normalized input; for $m = 2$ each coordinate polynomial has degree at most 2 because squaring is $\mathbb{F}_2$-linear. Left-chain encodings of $m \ge 3$ summands add $(m-2)n$ auxiliary Boolean coordinates.

3 The chain subsystem

Fix $m, k \ge 1$, let $r = mk$ and index the chain variables by $v \in \{0, \dots, r-1\}$ with row $i(v) = \lfloor v/k \rfloor$ and depth $j(v) = v \bmod k$. Let $G_{m,k}$ be the set of chain equations.

Lemma 1. *The graded-lex leading monomial of each chain equation is u_v^2. The standard exponents of $G_{m,k}$ are exactly the squarefree ones, so every standard exponent has degree at most mk.*

Proof. The tail $u_{v'}$ or 1 has degree at most one, so the leading term is u_v^2 (`leading_chainPolynomial`). An exponent e is standard exactly when no u_v^2 divides u^e, i.e. $e_v \le 1$ for all v; its degree is then at most r (`standard_le_one`, `standard_degree_bound`). □

Theorem 2 (Chain subsystem)**.** *Let K be a field with $2 \neq 0$. If $p \in \langle G_{m,k} \rangle$ and every monomial of p is squarefree, then $p = 0$.*

Proof. Let $\psi : K[u_v] \to K[z_1, \dots, z_m]$ be the K-algebra map $u_v \mapsto z_{i(v)}^{2^{j(v)}}$. For a chain equation, $\psi(u_v^2 - u_{v'}) = z_i^{2^{j+1}} - z_i^{2^{j+1}} = 0$ when $j+1 < k$ and $\psi(u_v^2 - 1) = z_i^{2^k} - 1$ when $j = k-1$ (`expand_chainPolynomial`). A squarefree monomial u^e with digit sets $S_i \subseteq \{0, \dots, k-1\}$ maps to $z^{\beta(e)}$ with $\beta(e)_i = \sum_{j \in S_i} 2^j < 2^k$ (`expand_monomial`, `bitExp_lt`); binary digits are determined by their weighted sum, so β is injective on squarefree exponents (`binary_sum_injective`, `bitExp_injective`).

Let L be an algebraic closure of K. Since $2^k \neq 0$ in K, the polynomial $X^{2^k} - 1$ is separable and splits in L, so its root set $S \subset L$ has exactly 2^k elements. Map $\psi(p)$ to $L[z]$. Every image of a chain equation vanishes at every point of S^m, hence so does the image of $p \in \langle G_{m,k} \rangle$. Its partial degrees are below $|S|$, so Mathlib's grid vanishing theorem (`eq_zero_of_eval_zero_at_prod_finset`) gives $\psi(p) = 0$ over L. The coefficient of $z^{\beta(e_0)}$ in $\psi(p)$ is the coefficient of u^{e_0} in p by injectivity of β, and the coefficient map $K \to L$ is injective, so every coefficient of p vanishes (`reduced_chain_ideal_trivial`). □

No Gröbner basis, normal form, closure rank or solution count is assumed in the proof; in particular the actual summation polynomial plays no role, which is why the bound holds for every input list.

Theorem 3 (Chain bound)**.** *Let K be a field with $2 \neq 0$, $m, k \ge 1$, and $F \supseteq G_{m,k}$ a set of polynomials of degree at most δ. Then* $\mathrm{LFD}(F) \le \max(mk + 1, \delta)$.

Proof. Lemma 1, Theorem 2 and the finite-quotient theorem with $s = mk$ (`chain_lfd_bound`). □

Corollary 4 (Two summands, every depth)**.** *For every prime $p \ge 5$, depth $k \ge 2$ and $a, b, t \in \mathbb{F}_p$, the normalized two-summand chain presentation $\{u_v^2 - tail\} \cup \{\mathrm{NF}(S_3)\}$ of `PrimeM2General` has* $\mathrm{LFD} \le 2k + 1$.

Proof. The chain equations of `PrimeM2General` are exactly $G_{2,k}$ up to sign of the tail (`chainInputs_two_subset`, a case analysis on $v \bmod k$), and every input has degree 2 (`PrimeM2General.degree_and_top`); apply Theorem 3 with $\delta = 2$ (`prime_m2_lfd_bound`, `prime_field_m2_lfd_bound`). □

Theorem 5 (No constant at degree three, every depth)**.** *Let K be a field with $2 \neq 0$, $k \ge 2$, and $a, b, t \in K$ with $3t^2 + a \neq 0$. For the normalized two-summand chain presentation F of Corollary 4, $1 \notin V_{F,3}$. In particular, if $\langle F \rangle$ is the unit ideal then* $\mathrm{LFD}(F) \ge 4$.

Proof. The inputs are quadratic tops q_i (the variable squares and the top $\alpha u_0 w_0 + \beta(u_0 w_1 + u_1 w_0) + u_1 w_1$ of g, with $\alpha = -2(t^2 + a)$ and $\beta = -2t$) plus tails of degree at most 1. By `QuadraticNoLow.no_one_at_three` it suffices that the tops have no homogeneous syzygy of degree 2 or 3 outside the trivial ones; the degree-2 statement is the pivot argument of `HomogeneousFirstFall`, and the degree-3 statement is `degree_three_syzygy_zero`, whose hypothesis $\alpha - \beta^2 \neq 0$ reads $-2(3t^2 + a) \neq 0$ (`PrimeM2Lower.no_one_at_three`, `lfd_ge_four_of_unit`). □

This generalizes the lower half of the all-depth family of `PrimeM2Sharp` ($b = t = 0$, where the condition is $a \neq 0$) to every curve and target. The condition $3t^2 + a \neq 0$ is the one that separates ordinary first fall degree 3 from 4 (`PrimeM2General.exact_ordinary_ffd`, `PrimeFieldConclusions.ordinary_ffd`), and the four sweep systems with $\dim V_3 \neq 2k+1$ are consistent with it.

Lemma 6 (Chain degree of the normalized summation polynomial; written)**.** *For $m \geq 2$ summands the normalized generator $\mathrm{NF}(S_{m+1}(x_1, \ldots, x_m, t))$ has chain degree at most $m \min(k, m-1) \leq mk$.*

Proof. S_{m+1} has degree 2^{m-1} in each x_i. A monomial $\prod_i x_i^{e_i}$ with $e_i \leq 2^{m-1}$ reduces by the chains to the squarefree chain monomial with digit set the binary digits of $e_i \bmod 2^k$, of weight at most $\min(k, \mathrm{wt}(e_i)) \leq \min(k, m-1)$, since $e_i \leq 2^{m-1}$ has at most $m-1$ binary digits (the value 2^{m-1} itself has one). Summing over the m blocks gives the bound. Normalization only reduces exponents modulo 2^k and collects terms, which cannot increase the weight. □

Corollary 7 (Unconditional dimension bound)**.** *For every $m \geq 2$, $k \geq 1$, prime $p \geq 5$ with $2^k \mid p-1$, curve and target, the direct chain presentation $\{\mathrm{NF}(S_{m+1})\} \cup G_{m,k}$ has* LFD $\leq mk+1$.

Proof. A normal form is a polynomial in the reduced space of the chains, whose monomials are squarefree in mk variables, so it has degree at most mk; with the chain equations of degree 2, Theorem 3 applies with $\delta = \max(2, mk)$. This is the kernel-checked `ChainQuotient.reduced_inputs_lfd_bound`, which takes any inputs in the reduced space; Lemma 6 only sharpens the degree of the normalized summation polynomial and is not needed for the bound. □

Remark 8. The hypothesis $2 \neq 0$ is used only to obtain 2^k distinct roots of unity. Buchberger's coprime-leading-term criterion shows that $G_{m,k}$ is a Gröbner basis in every characteristic, so the statement holds in characteristic two as well, but that route is not formalized here.

4 One bound per family

Table 1 lists the kernel-checked statement for each family. The binary entries are the descended left-chain S3 systems of `BinaryS3Bounds`, which allow every target and every S3 incidence graph; the Koblitz entry is the specialization $b = 1$ and is identical in content, because the bound uses only the Boolean field equations and the input degrees. The Frobenius structure of a symmetric factor base therefore cannot raise the bound, and the sweep below tests whether it changes the measured value.

The bound $mk+1$ is the dimension bound of the finite quotient and is attained by abstract Boolean systems and, for $m \geq 3$ summands at depth $k \leq m-1$, by the summation systems themselves (Section 7). For the measured two-summand curve systems the exact values are much smaller; the written uniform selector theorem of the companion manuscript gives $\min(k, \ell) + 2$ for binary two-summand systems on spaces of dimensions k, ℓ, and the inverse family shows that linear growth in k does occur. The same mechanism gives a written bound for prime chains.

Theorem 9 (Selector bound; written proof)**.** *Let $p \geq 5$ be prime, $k \geq 2$ with $2^k \mid p-1$, and $a, b, t \in \mathbb{F}_p$ with $4a^3 + 27b^2 \neq 0$. The normalized two-summand chain presentation $G_{2,k} \cup \{\mathrm{NF}(S_3(x_1, x_2, t))\}$ over $\mathbb{F}_p$, for the subgroup base and for every coset base, satisfies* LFD $\leq k+2$.

Proof sketch. Work in the chain quotient $\mathbb{F}_p[(\mathbb{Z}/2^k)^2]$ with Hamming-weight degree; the passage to the ordinary ring is Theorem 12. For a point α of the first block, the character

Family	Presentation	Bound	Lean statement
Prime	m chains of depth k, inputs of degree $\le \delta$	$\max(mk+1, \delta)$	`prime_chain_lfd_bound`
Prime	$m = 2$, normalized S_3, every $k \ge 2$, $p \ge 5$	$2k+1$	`prime_m2_lfd_bound`
Binary	descended left-chain S_3, $r \ge 2$ Boolean coordinates	$r+1$	`binary_descent_lfd_bound`
Koblitz	the same with $b = 1$	$r+1$	`koblitz_descent_lfd_bound`
Boolean sharpness	$\{Z_i^2 - Z_i\} \cup \{\text{one degree-}r \text{ row}\}$, $r \ge 2$	$= r+1$	`BooleanSharp.exact_lfd`

Table 1: Kernel-checked bounds; the first four statements are in the `CurveBounds` namespace. For m summands on coordinate spaces of dimension k, $r = mk$ (plus $(m-2)n$ auxiliary coordinates in a left chain), so every family has the bound $mk+1$ for the direct encodings without auxiliary coordinates, $2k+1$ for two summands. The sharpness row is an abstract Boolean family, not a curve system.

Family ($m = 2$)	Proved upper	Proved lower (family)	Written	Measured, certified
Prime chains, depth k	$2k+1$; $mk+1$ for all m (Lean)	≥ 4 for unit ideals with $3t^2 + a \ne 0$, every k; $= 4$ for $y^2 = x^3 + ax$, target $(0,0)$, $a \ne 0$, $a^{2^k} \ne 1$ (Lean)	$\le k+2$, nonsingular curve (written, Theorem 9)	4 ($k \le 5$); 5 ($k = 6, \dots, 9$, generic targets); 6 ($k = 10, \dots, 13$, unit instance over $p = 65537$); 7 ($k = 14$, upper bound by the degree lift)
Binary, spaces of dimension k, ℓ	$k + \ell + 1$ (Lean)	—	$\le \min(k, \ell) + 2$; worst case over dimension $\le k$ at least $\lfloor k/2 \rfloor + 1$ for $k \ge 6$ (inverse family; for odd k the witness has dimension $k-1$)	≤ 5 on the grids through $k = 10$; 5 occurs from $k = 8$
Koblitz, same	$k + \ell + 1$ (Lean)	—	as binary (the inverse family is Koblitz)	≤ 5 on the grids through $k = 10$; 5 occurs from $k = 8$
Boolean abstract, r coordinates	$r+1$ (Lean)	$= r+1$ (Lean)	—	—

Table 2: Two-sided summary. "Lean" entries are kernel-checked (`PrimeFieldConclusions.sharp_lfd`, `BooleanSharp.exact_lfd` for the lower rows); "written" entries are proved in the companion manuscript, in Theorem 9 or in Lemma 6; measured values are exact and certified on the seeded grids of Sections 5–6.

idempotent e_α is a polynomial in the k chain variables of that block, of weight k, and $e_\alpha g$ has weight at most $k+2$ and equals e_α times the fibre polynomial $g(\alpha, x_2)$, of degree at most 2 in x_2. Nonsingularity of the curve guarantees that no fibre polynomial vanishes identically, so on each fibre the functions on the zero set are represented by ρe_α with $\rho \in \{1, x_2\}$, of weight at most $k+1 = D-1$ for $D = k+2$. These representatives span a complement of the ideal, with projector π. Using the chain system of the fibre polynomial (every degenerate case included: constant, linear, double root, roots outside the subgroup), each correction $v \cdot \rho e_\alpha - \pi(v \cdot \rho e_\alpha)$ for a chain variable v, and $1 - \pi(1)$, lies in V_{k+2}. Induction on monomials then gives $m - \pi(m) \in V_{\max(k+2, \mathrm{wt}\, m)}$ for every monomial m, which is the hypothesis of the local-to-global criterion (`LocalReductionBound.lfd_bound_of_local`, whose conditions `bounded` and `room` are the weight bound $k+1$ of the representatives), and it gives the bound. The full proof is in the repository (Section 11). □

The theorem is not formalized; the kernel-checked bound remains $2k+1$, which needs no hypothesis on a, b, t. A separate review checked every step of the proof, and the statement was tested by exact computation in the ordinary ring, with an implementation independent of the proof, on 149,122 systems with $2 \le k \le 5$, including every nonsingular (a, b, t) over $\mathbb{F}_5$, $\mathbb{F}_{13}$, $\mathbb{F}_{17}$, $\mathbb{F}_{29}$, $\mathbb{F}_{37}$ at $k = 2$. This test has no working negative control: random degree-2 generators exceed the bound only at unequal depths, and at equal depths an independent check of 292 random generators during review (`docs/reviews-20261112/referee-U1.md`) found none above $k+2$, so a failure of the test was not demonstrated to be possible. The per-instance records of the test were not kept; only its counts are recorded, in `docs/reviews-U1-upper-bound-redteam-20261110.md`. The bound is attained at $k = 2$, where LFD $= 4$ exactly when $3t^2 + a \neq 0$ and 3 otherwise on all 132,303 systems. For $k \ge 3$ no computed system attains it (the largest exact values are 4 for $k \le 5$, 5 for $k = 6, \ldots, 9$ and 6 at $k = 10$), so it is not claimed to be sharp. For singular (a, b) the proof does not apply; the measured values there are still at most $k+2$ on 2570 systems. The method cannot give a slope below 1 in k: any localisation at one value of x_1 already has weight k.

5 Sweep design

The sweep is generated deterministically from a fixed seed. Every case records the field, the curve, the depth or dimension k, the factor-base class, the target and its class, the number of scalar S_3 solutions, the number of rational decompositions checked with the group law, and the hash of the exact input system.

Prime curves. Primes $97, 193, 257, 641, 7681$ with $2^k \mid p-1$, $k = 2, \ldots, \min(7, v_2(p-1))$; four curves per prime (the $j = 0$ and $j = 1728$ curves when nonsingular, and random (a, b)); factor bases *subgroup* (H itself; symmetric: closed under negation and under multiplication by H) and two *random cosets* cH with $c \notin H$ (absent only when $H = \mathbb{F}_p^*$); targets: a point of order two when the cubic has a root, two random rational points, and one *planted* target $P + Q$ with P, Q in the base, which guarantees at least one rational decomposition.

Binary ordinary and Koblitz curves. $n = 5, \ldots, 12$; curves $(a, b) = (0, 1), (1, 1)$ (Koblitz) and two random ordinary curves with $b \neq 1$; $k = 2, \ldots, \min(7, n-1)$; factor bases: up to two *Frobenius-invariant* subspaces of dimension k (symmetric; the subfield $\mathbb{F}_{2^k}$ when $k \mid n$ is preferred) enumerated as the images of $g(\sigma)$ for the divisors g of $X^n - 1$ over $\mathbb{F}_2$, and two *random subspaces* that are not Frobenius-invariant; targets as in the prime case, with the order-two point $(0, \sqrt{b})$.

Depths 8 and 9. The dense chain closure does not reach $k = 8$ (its product step materializes all variable multiples of every low row). A second engine uses the group-algebra structure of the chain quotient: multiplication by a chain variable adds 2^j to a block exponent modulo 2^k, so a product is a permutation of a row's coefficients. It computes the exact dense profile through degree 4, then builds the degree-5 row space from the normalized multiples of the generator and the $2k$ shifts of every echelon row whose pivot has weight below 5, each low pivot processed once. Every row lies in V_5 by construction; processing each low pivot once is a heuristic schedule, and exactness, when claimed, rests on the certificate or on the final verification pass. Reaching rank $|M_{\leq 5}| - |Z|$ with the evaluation of $M_{<5}$ onto Z certifies $V_5 = I \cap M_{\leq 5}$, and the exact dimension of V_4 decides the strict fall at 5. There are 20 systems at $k = 8$ (primes 257, 769, 7681; $y^2 = x^3 + x$ and a random curve; subgroup and random coset; random-rational and planted targets) and two $k = 7$ controls, and 8 systems at depth 9 over $p = 7681$ (same curves, bases and target classes) with two further $k = 7$ controls; at $k = 9$ the degree-5 closure has 12,616 columns.

Certification. Binary systems use the Boolean closure with containing-space checks and the local-to-global rank certificate: once the evaluation of the monomials of degree below D is onto the solution set and the closure quotient at D has codimension equal to the number of solutions, every polynomial of degree at most D reduces at budget D to a representative of lower degree, so the full ideal has LFD $\leq D$ (`LocalReductionBound.lfd_bound_of_local`) and the exact value is the last strict fall. Prime systems use the exact chain closure modulo the chains with the independent exclusion certificate at every degree and the same rank criterion: the chain ideal is radical with all 2^{mk} points rational, so the full ideal is the vanishing ideal of the scalar solution set, and the squarefree monomials evaluate at a solution (E_1, E_2) as $g^{a_1 E_1 + a_2 E_2}$ for the subgroup generator g. Unit ideals (no solution) are exact at the first degree deriving 1. Degrees are scanned through 6 and all matrices are exact; a case that is not certified by then is reported as capped with its interval and never as a value.

6 Results

The sweep has 3163 cases and 3005 distinct systems; 3005 systems have a certified exact LFD and 0 are capped. The histogram of exact values over all distinct systems is 0: 66, 1: 11, 2: 625, 3: 774, 4: 1217, 5: 312 (None = capped). All exact values satisfy the kernel-checked bound $2k + 1$; the binary values also satisfy the written bound $k + 2$.

k	subgroup (symmetric)	random coset	bound $2k$+1
2	4:75 (max 4)	3:3, 4:142 (max 4)	5
3	4:78 (max 4)	4:157 (max 4)	7
4	4:78 (max 4)	4:157 (max 4)	9
5	4:79 (max 4)	4:158 (max 4)	11
6	4:4, 5:59 (max 5)	4:8, 5:118 (max 5)	13
7	4:3, 5:45 (max 5)	4:6, 5:90 (max 5)	15

Table 3: Prime curves: distinct systems by exact LFD (value:count) for each factor-base class and depth or dimension k, with the number of capped systems and the maximum certified value.

Random versus symmetric bases. Prime: over 311 cells (field, curve, k, target class) with both classes certified, the maximum exact LFD of the random bases exceeds that of the symmetric bases in 0 cells, is lower in 0 and equal in 311; the cell means are 4.24 (random)

k	Frobenius-invariant (symmetric)	random subspace	bound $2k{+}1$
2	0:12, 1:5, 2:28, 3:2 (max 3)	0:23, 1:3, 2:83, 3:2 (max 3)	5
3	0:8, 2:36, 3:11 (max 3)	0:7, 2:83, 3:34 (max 3)	7
4	0:3, 2:30, 3:35, 4:3 (max 4)	2:36, 3:91 (max 3)	9
5	0:1, 2:22, 3:13, 4:4 (max 4)	3:101, 4:11 (max 4)	11
6	0:2, 2:8, 3:26, 4:11 (max 4)	3:70, 4:26 (max 4)	13
7	2:8, 3:10, 4:14 (max 4)	3:20, 4:60 (max 4)	15

Table 4: Binary ordinary curves: distinct systems by exact LFD (value:count) for each factor-base class and depth or dimension k, with the number of capped systems and the maximum certified value.

k	Frobenius-invariant (symmetric)	random subspace	bound $2k{+}1$
2	0:4, 1:1, 2:26, 3:6 (max 3)	0:2, 1:1, 2:87, 3:1 (max 3)	5
3	0:3, 1:1, 2:23, 3:17 (max 3)	0:1, 2:78, 3:29 (max 3)	7
4	2:19, 3:34, 4:6 (max 4)	2:32, 3:79 (max 3)	9
5	2:11, 3:16, 4:6 (max 4)	3:85, 4:11 (max 4)	11
6	2:8, 3:17, 4:17 (max 4)	3:56, 4:28 (max 4)	13
7	2:7, 3:4, 4:17 (max 4)	3:12, 4:58 (max 4)	15

Table 5: Koblitz curves: distinct systems by exact LFD (value:count) for each factor-base class and depth or dimension k, with the number of capped systems and the maximum certified value.

and 4.24 (symmetric). Binary ordinary: over 164 cells (field, curve, k, target class) with both classes certified, the maximum exact LFD of the random bases exceeds that of the symmetric bases in 34 cells, is lower in 10 and equal in 120; the cell means are 2.84 (random) and 2.64 (symmetric). Koblitz: over 164 cells (field, curve, k, target class) with both classes certified, the maximum exact LFD of the random bases exceeds that of the symmetric bases in 9 cells, is lower in 42 and equal in 113; the cell means are 2.87 (random) and 3.06 (symmetric).

Growth in k and the gap to the bound. Prime: $k = 2$: max 4, mean 3.99, bound 5; $k = 3$: max 4, mean 4.00, bound 7; $k = 4$: max 4, mean 4.00, bound 9; $k = 5$: max 4, mean 4.00, bound 11; $k = 6$: max 5, mean 4.94, bound 13; $k = 7$: max 5, mean 4.94, bound 15. Binary ordinary: $k = 2$: max 3, mean 1.53, bound 5; $k = 3$: max 3, mean 2.08, bound 7; $k = 4$: max 4, mean 2.64, bound 9; $k = 5$: max 4, mean 2.93, bound 11; $k = 6$: max 4, mean 3.16, bound 13; $k = 7$: max 4, mean 3.59, bound 15. Koblitz: $k = 2$: max 3, mean 1.95, bound 5; $k = 3$: max 3, mean 2.24, bound 7; $k = 4$: max 4, mean 2.74, bound 9; $k = 5$: max 4, mean 3.05, bound 11; $k = 6$: max 4, mean 3.29, bound 13; $k = 7$: max 4, mean 3.69, bound 15.

These are exact values of the specified presentations on a seeded grid. They support no population inference, no statement about other bases, and no relation-cost claim; capped systems retain only the interval from their last strict fall to the proved bound.

Observations. All statements here concern the seeded grids. First, for prime chains the exact LFD is 4 at every $k \leq 5$ except three systems at $k = 2$ with value 3 (two unit ideals and one proper ideal with two solutions, all with $3t^2 + a = 0$), and 5 at $k = 6$ and 7 except 21 systems with value 4 (12 at $k = 6$, 9 at $k = 7$), all on the $j = 1728$ curve $y^2 = x^3 + x$ with the target $(0, 0)$ of order two. For unit ideals in this family `PrimeM2Sharp` proves exact LFD 4 at every depth when $a \neq 0$ and $a^{2^k} \neq 1$; the sweep members have $a = 1$, so for the subgroup base they are proper ideals, measured and not covered by the kernel statement. The

same primes give 4 at $k = 5$. The depth-8 runs find exact LFD 5 with strict falls at 4 and 5 in all 20 systems (12 subgroup, 8 coset; 19 proper, one unit ideal) and the depth-9 runs in all 8 systems (4 subgroup, 4 coset, 24 to 44 solutions, one prime); these runs contain no order-two targets, and the four $k = 7$ controls (fresh targets on $y^2 = x^3 + x$, $p = 7681$) give the sweep's generic value, LFD 5 with falls at 4 and 5. At depth 10, LFD ≥ 6 holds for the three instances of Section 7, with exact value 6 on the unit instance (Table 6); on the unit instance over $p = 65537$ of that section the exact value is 6 at $k = 10, \ldots, 13$ and 7 at $k = 14$, the upper bound at $k = 14$ by the degree lift described there. Second, for prime curves the random coset and the subgroup agree in every paired cell. This comparison does not test symmetry: since S_3 is weighted homogeneous, the coset system cH is the subgroup system of a curve with rescaled coefficients (a twist when c is a non-square), and cH is itself closed under negation and under multiplication by H. Third, on binary curves the two base classes differ in the raw comparison and the difference follows the solution structure. We split the symmetric class into subfields and Frobenius-invariant non-subfield spaces and compare paired cells (field, curve, k, target class) with a two-sided sign test on the discordant cells. On ordinary curves random subspaces have a larger maximum than invariant non-subfield spaces in 22 of 30 discordant cells ($p \approx 0.016$); over proper ideals only 3 against 5 ($p \approx 0.73$), so the difference is carried by unit ideals (the subfield class is 70% unit ideals with about one solution). On Koblitz curves the invariant non-subfield class has the larger maximum in 31 of 38 discordant cells and in 20 of 22 over proper ideals ($p \approx 1.2 \cdot 10^{-4}$); these bases have twice the mean number of solutions (13.2 against 6.4; the solution set is Frobenius-stable) and the LFD grows with the solution count in every class. Within strata of equal k and solution-count bucket the invariant class is lower on average (-0.18 Koblitz, -0.28 ordinary; sign tests 6:12, $p \approx 0.24$, and 5:13, $p \approx 0.10$). The grid is consistent with the effect of invariance being mediated by the solution count, which is an association on small strata, not a causal decomposition.

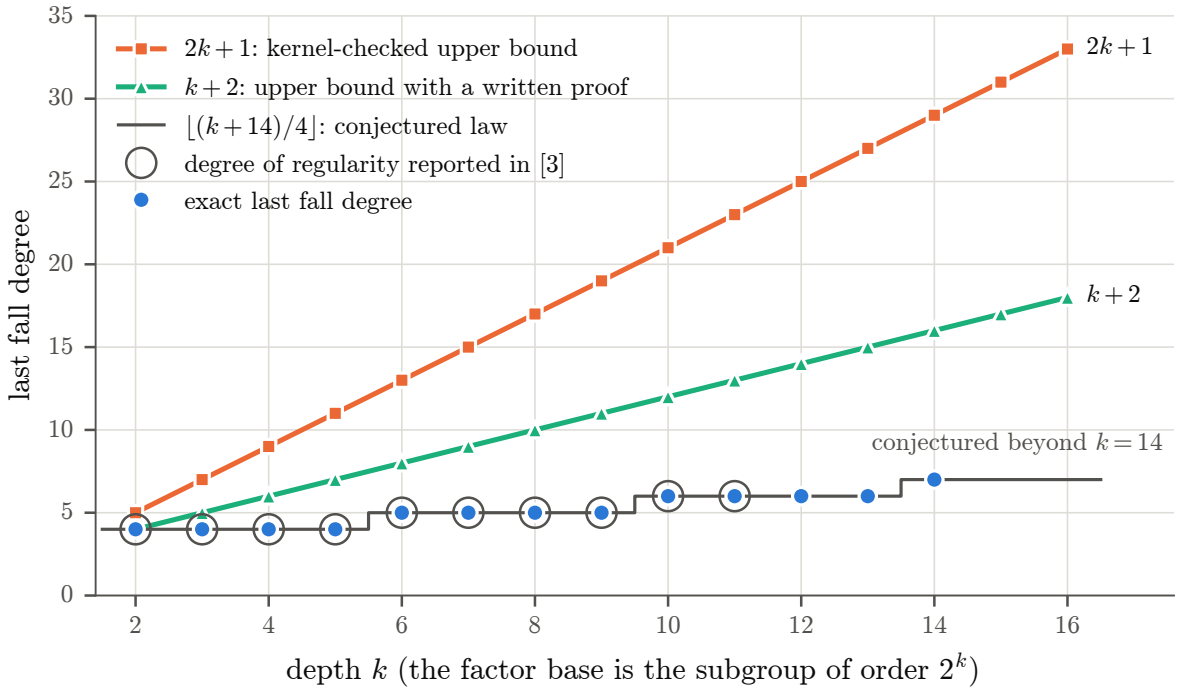


Figure 1: The two-summand prime chain profile against the proved bounds. Dots: exact LFD on the instances computed (Table 6; the upper bound at $k = 14$ is by the degree lift). Rings: the averaged degree of regularity of Petit, Kosters and Messeng [3, Table 4]. The step line is the conjectured law $\lfloor (k + 14)/4 \rfloor$.

k	exact LFD	$\dim V_5$	V_6	evidence	PKM
2–5	4 (3 for three systems at $k=2$ with $3t^2+a=0$)	—	—	sweep; local-to-global, containing space	4
6, 7	5 (4 on 21 systems, $y^2=x^3+x$, target $(0,0)$)	—	—	sweep; local-to-global; one $k=6$ instance exact in Lean (Theorem 14)	5
8	5	—	—	20 systems; shift engine, rank certificate $V_5 = I \cap M_{\leq 5}$	5
9	5	—	—	8 systems; shift engine, rank certificate; recomputed	5
10	6 (≥ 6 on two more instances)	18,569	full	unit instances over $p=12289$ (two engines) and $p=65537$; closed V_5 without 1; $1 \in V_6$	6
11	6	25,033	full	unit instance, $p=65537$ (third engine); ≥ 6 on two instances, $p=12289$	6
12	6	32,057	full	as $k=11$	—
13	6	39,657	full	unit instance, $p=65537$; ≥ 6 on two instances, $p=40961$; random curve, $p=65537$: $V_6 = I \cap M_{\leq 6}$	—
14	7 (≥ 7 certified; ≤ 7 by the lift, uncertified)	47,849	closed, $1 \notin V_6$	unit instances, $p=65537$ and 163841, random curves, $p=65537$ and 114689: $\dim V_6 = 448{,}922$; independent checker on three; lift on three: V_7 is the degree-7 part of the ideal	—
15	≥ 7	56,649	$1 \notin V_6$	unit instance, $p=65537$; Theorem 15	—
every $k \geq 2$	≥ 4 if $3t^2+a \neq 0$, unit ideal			Lean (Theorem 5)	—
every $k \geq 2$	$\leq 2k+1$			Lean (Corollary 4)	—

Table 6: The two-summand prime chain profile. Rows 2–15 are certified exact values or bounds on the computed instances (Sections 6 and 7); the last two rows are kernel-checked for every instance; the upper bound in row 14 is the degree lift of Section 7, which has a written proof and no certificate. The columns $\dim V_5$ and V_6 give the closures of the unit instance over $p=65537$ for $k \geq 10$; "full" means $V_6 = M_{\leq 6} \ni 1$, so that the unit ideal is derived at degree 6; dashes: not needed, since the value is decided at degree 4 or 5. The last column is the corrected degree of regularity that Petit, Kosters and Messeng report for the same subgroup-chain presentation with Magma, averaged over 100 random instances per depth ($n_1 = k$, about one solution per instance) [3, Table 4].

An empirical account of the step at depth six. The exact dense closures V_3 and V_4 of two-summand chain systems were computed for $k = 3, \dots, 10$ on five curves over $p \in \{257, 7681, 12289\}$ (all five through $k = 8$, four at $k = 9$, two at $k = 10$) with random rational targets, and are recorded for every prime system of the sweep. On the five depth-step curves $\dim V_3 = 2k + 1$; in the sweep this fails only for three systems at $k = 2$ and one at $k = 6$, consistent with the kernel-checked Theorem 5 above, which excludes 1 from V_3 whenever $3t^2 + a \neq 0$. For $k \leq 5$, V_4 is the whole degree-4 part of the ideal, $\dim V_4 = |M_{\leq 4}| - |Z|$, so the certificate holds at degree 4. For $k \geq 6$ on the depth-step curves, $\dim V_4 = 4k^2 + 172k - 571$ exactly at $k = 6, \dots, 10$ (605, 829, 1061, 1301, 1549), with $\dim(V_4 \cap M_{\leq 3}) = 4k + 87$, independent of the prime, the curve and the number of solutions among these instances. This is not universal: 86 of the 333 prime systems at $k = 6, 7$ in the sweep deviate, mostly order-two targets (607 and 835 instead of 605 and 829), the $(0, 0)$-target $j = 1728$ systems (730, 794, 1343, 1471; $1 \in V_4$ when $a^{2^k} \neq 1$, kernel-checked), one tangent target with $3t^2 + a = 0$ (603) and one target of the class C_2 of Section 7 (857). On the generic instances the quadratic dimension meets the quartic $|M_{\leq 4}| = \binom{2k}{\leq 4}$ between $k = 5$ ($389 \geq 386$, extrapolated) and $k = 6$ ($605 < 794$): beyond it the degree-4 closure cannot contain the degree-4 part of the ideal and LFD ≥ 5; the matching upper bound 5 is the separately measured certificate at degree 5 for $k \leq 9$, and it fails at $k = 10$. For $k \geq 8$ the identity is bounded above per instance: $V_4(k)$ lies in the free high-level extension of $V_4(7)$, of dimension $829 + 230(k - 7) + 2\binom{2k-14}{2} = 4k^2 + 172k - 571$ (a written count), by the kernel-checked reduction theorem (Theorem 13), which turns closedness at depth 8 into closedness at every depth $k \geq 8$. Equality is a computation: it was checked at $k = 8, \dots, 12$ on one instance and at $k = 8, 9$ on two more, and $\dim V_4$ was confirmed through $k = 13$ by an independent engine. What remains open is the genericity of the depth-7 base in (a, b, t); with it, LFD ≥ 5 would follow for every such system with $|M_{\leq 4}| - |Z|$ above the bound.

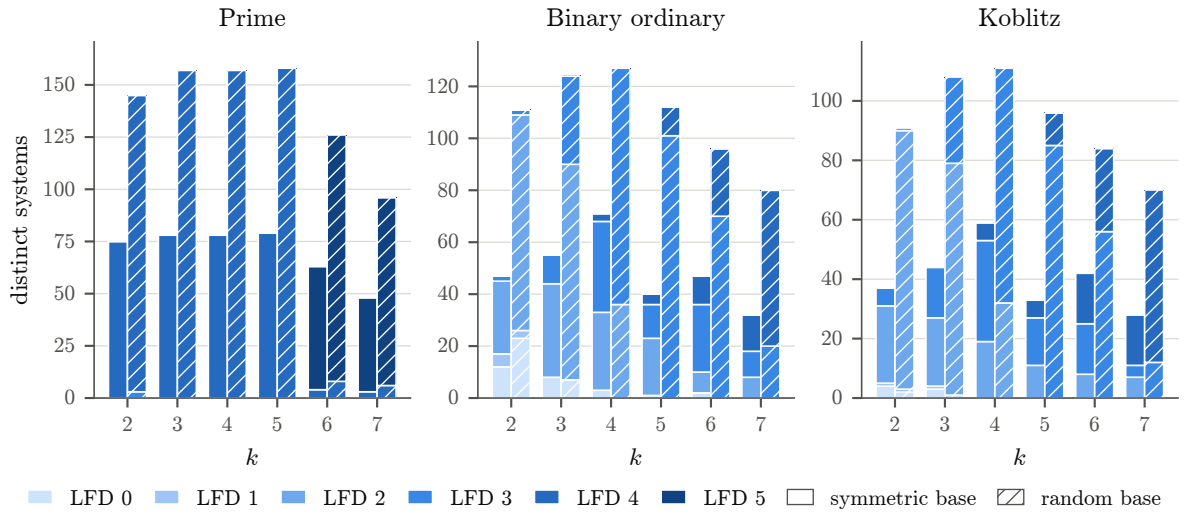


Figure 2: Exact LFD in the sweep, by depth or dimension k, for each family (stacked counts of distinct systems). In each pair the left bar is the symmetric factor base (the subgroup, or a Frobenius-invariant subspace) and the hatched right bar the random one (a random coset, or a random subspace).

7 What a tight upper bound would need

The proved bounds are dimension bounds, linear in k, while every certified two-summand value is 3, 4 or 5 for prime chains through $k = 9$ (with LFD = 6 on a unit instance at $k = 10, \dots, 13$, LFD = 7 on it at $k = 14$, and LFD ≥ 6 on every other instance computed at $k = 10, \dots, 13$) and at most 5 through $k = 10$ for the binary and Koblitz systems computed, which exclude the inverse family with value $\lfloor k/2 \rfloor + 1$. This section states, as precisely as the evidence allows, what separates the two.

The mechanism, restated in the group algebra. Modulo the chains, the two-summand prime presentation lives in the group algebra $A = K[G]$, $G = (\mathbb{Z}/2^k)^2$: a squarefree chain monomial is a pair of block exponents, its degree is the Hamming weight $\mathrm{wt}(a_1) + \mathrm{wt}(a_2)$, and multiplication by a chain variable adds 2^j to one block exponent. The normalized generator g has weight at most 2, the ideal is $I = gA$, and because the chain ideal is radical with all 4^k points rational, I is the vanishing ideal of the scalar solution set Z. The closure V_D is the smallest subspace of $M_{\leq D}$ containing the shifts of g by exponent patterns of weight at most $D - 2$ and closed under shifting its elements of weight below D by 2^j; shifts that carry lower the weight. Carries are not the only source of strict falls: the same generator closed in the carry-free algebra $K[u]/(u^2)$ still has a fall at degree 4. The certificate of Section 5 says LFD $\leq D$ as soon as $\dim V_D = |M_{\leq D}| - |Z|$ and $M_{<D}$ evaluates onto Z.

Generic closure dimensions. Write $d_D(k)$ for the dimension of the degree-D closure of a generic two-summand chain system, the value attained on a Zariski-open set of coefficient vectors of g, if such a value exists. Section 6 shows that it cannot be the value for every instance: order-two targets and the $(0, 0)$-target $j = 1728$ family have larger $\dim V_4$, and the latter contains 1 in V_4 at every depth. On the instances we regard as generic (random curves and random targets of large order), the data are consistent with $d_3(k) = 2k + 1$, $d_4(k) = 4k^2 + 172k - 571$ for $k = 6, \dots, 10$, and $d_5(k) \geq |M_{\leq 5}| - |Z|$ for every $k \leq 9$, where the degree-5 closure has not yet separated from the ideal. For $k = 10, \dots, 16$ the computed values are $d_5 = 18{,}569$, 25,033, 32,057, 39,657, 47,849, 56,649, 66,073, and on the unit instance over $p = 65537$ the degree-6 closure is $M_{\leq 6}$ for $k \leq 13$ and has dimension $448{,}922 < |M_{\leq 6}|$ at $k = 14$, where the degree-7 closure is $M_{\leq 7}$.

The conjectural formula. If the closures of an instance attain the generic values, the local-to-global certificate gives an upper bound at the first degree where they meet the ideal, and the dimension count a lower bound below it:

$$\mathrm{LFD}(k) = \min\{D : \ d_D(k) = |M_{\leq D}| - |Z| \text{ and } M_{<D} \to K^Z \text{ is onto}\},$$

up to the strict-fall rule for the exact value. Every tight bound for such instances is then a statement about the functions d_D. Conjecturally, if d_D is a polynomial of degree below D, it cannot keep pace with $\binom{2k}{\leq D}$ and a step beyond D occurs at the first depth k_0 at which $d_D(k_0) < \binom{2k_0}{\leq D} - |Z|$. The first such depths computed are $k_0 = 2, 6, 10, 14$ for $D = 3, 4, 5, 6$, that is $4D - 10$ (Figure 3), and d_5 is a cubic on the computed range (the recursion below).

The depth-ten measurement. We computed instances at $k = 10$. The first is $p = 12289$, $y^2 = x^3 + x$ (the $j = 1728$ curve), a random rational target (abscissa 9362, not of order two) whose two-summand system has no scalar solution, so its ideal is the unit ideal and $|M_{\leq 5}| - |Z| = 21{,}700$. The exact dense V_4 has dimension $1549 = d_4(10)$, matching the quadratic. The degree-5 shift closure, with every low pivot processed once, stops at rank 18,569 after 17 rounds; re-reducing all $2k = 20$ variable shifts of all 3163 echelon rows of weight below

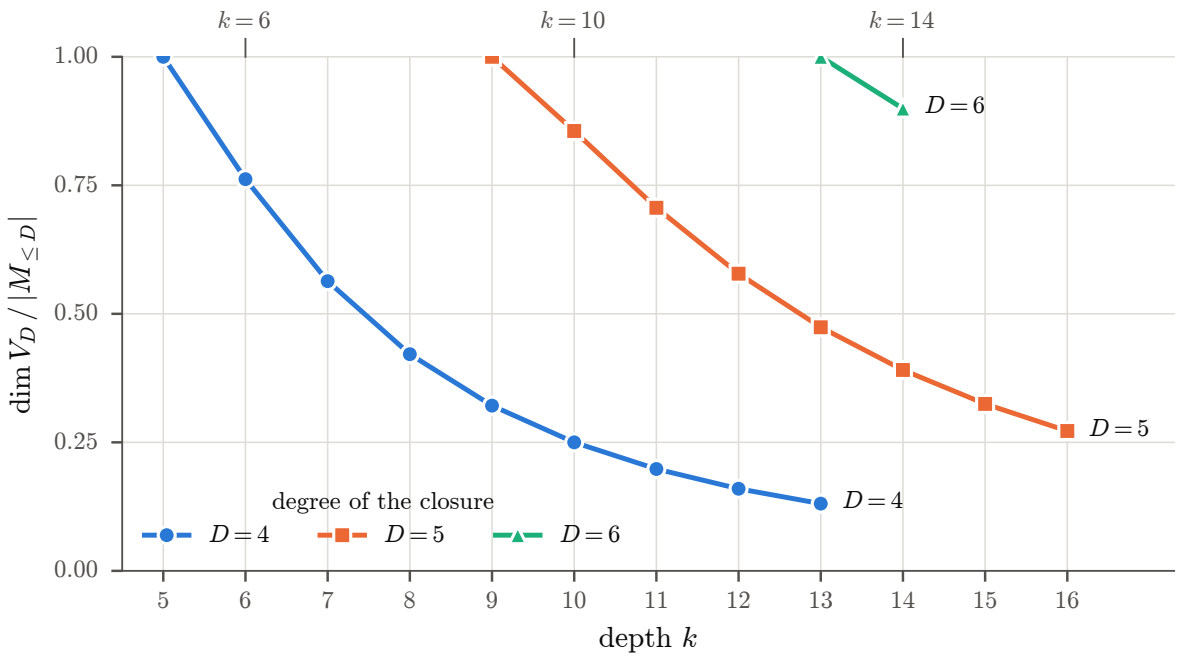


Figure 3: Share of the degree-D space $M_{\leq D}$ reached by the closure V_D on unit-ideal instances, for $D = 4, 5, 6$. Each curve starts at the last depth at which the closure is the whole space. The first deficient depths are 6, 10 and 14, that is $4D - 10$. The values for $D = 4$ are the identity $\dim V_4 = 4k^2 + 172k - 571$; those for $D = 5$ and $D = 6$ are the closures of this section (at $k = 16$ on the random curve).

5 against the final basis adds nothing, so the row space is closed under every admissible product and is exactly V_5. It has codimension 3131 in $M_{\leq 5}$ and does not contain 1. Since 1 lies in the ideal, LFD ≥ 6 for this system. The conclusion does not depend on the unit-ideal claim: the system has at most $2 \cdot 1024$ scalar solutions, so $\dim(I \cap M_{\leq 5}) \geq 21{,}700 - 2048 = 19{,}652 > 18{,}569$ in any case. For this instance $\dim V_5 = 18{,}569$; it equals $d_5(10)$ if the instance is generic. The degree-5 closure has separated from the ideal at $k = 10$, whereas it was full on every instance computed at $k = 9$; this is consistent with, not predicted by, the dimension account. An independent rerun from scratch is reported in Section 8. Two further instances give the same degree-5 closure. The second is a proper ideal on the same curve family, $p = 13313$, $y^2 = x^3 + x$, target abscissa 7412, with 98 scalar solutions; the third is a random curve, $p = 15361$, $y^2 = x^3 + 4021x + 9917$, target abscissa 7776, with 60 solutions. In both the verified-closed V_5 has rank 18,569 (targets 21,602 and 21,640), does not contain 1, and so LFD ≥ 6. All three instances have $\dim V_4 = 1549$ and the same symmetric and antisymmetric ranks $9311 + 9258$ of V_5 under the summand swap.

The exact value at depth ten. For the unit instance the degree-6 closure contains 1, so LFD $= 6$ exactly. The computation starts from the verified V_5 above, embeds it in the degree-6 space, splits it into its parts symmetric and antisymmetric under the summand swap σ (both lie in V_6, since the system is σ-invariant and 2 is invertible), and adds, in each round, random linear combinations of the $2k$ variable shifts of the rows whose leading monomial has weight below 6. Every added row is a product admitted by the closure rule, so every row lies in V_6 whatever the random choices; the certificate is one-sided and sound. After eight rounds both parts have full rank, $30{,}318 + 30{,}142 = 60{,}460 = |M_{\leq 6}|$, so $V_6 = M_{\leq 6} \ni 1$. The falls are visible in the ranks: after round seven the rank is 59,371, above the $18{,}569 + \binom{20}{6} = 57{,}329$ that a closure without a new element of weight below 6 can reach, and round eight shifts

exactly 18,569 + 2042 low rows. An independent engine, started from the generator alone, reaches the same closure in 453 rounds: $V_6 = M_{\le 6}$ with part ranks 30,318 + 30,142, $1 \in V_6$, and a deterministic sweep of all 21,700 low rows with no residual. With the lower bound this gives the exact profile 4 for $k = 2, \dots, 5$, 5 for $k = 6, \dots, 9$ and 6 at $k = 10$ on the instances computed, matching the averaged degree of regularity that Petit, Kosters and Messeng report for the same subgroup-chain presentation at every depth $k = 2, \dots, 10$, and at $k = 11$ [3, Table 4].

Depths eleven to fourteen. Two sets of closures extend the depth-ten measurement: closures of an independent engine at depths 11 to 13 over $p = 12289$ and $p = 40961$, and closures of a third engine on a GPU at depths 10 to 16 over $p = 65537$ and at depths 13 and 14 over $p = 114689$ and $p = 163841$. The independent engine (written without reference to the other engines of this paper, with a full deterministic closure sweep) reproduces $\dim V_5 = 18{,}569$ on all three depth-ten instances and computes $\dim V_5 = 25{,}033$ at $k = 11$ on two instances ($p = 12289$, a unit ideal on $y^2 = x^3 + x$ and a proper ideal with 348 solutions on $y^2 = x^3 + 4021x + 9917$), closed and without 1. Since $|Z| \le 2^{k+1}$, the degree-5 part of the ideal has dimension at least $\binom{22}{\le 5} - 4096 = 31{,}347 > 25{,}033$, so LFD ≥ 6 at $k = 11$ for both. The same engine gives $\dim V_4 = 4k^2 + 172k - 571$ on two instances at every $k = 6, \dots, 13$. At $k = 12$ the same engine gives $\dim V_5 = 32{,}057$ on both instances ($p = 12289$; $y^2 = x^3 + x$ with 1352 solutions and $y^2 = x^3 + 4021x + 9917$ with 1399), with identical profile $(0, 0, 2, 146, 3727, 32057)$ and part ranks 16,061 + 15,996, and at $k = 13$ ($p = 40961$, the same two curves, 1484 and 1582 solutions) $\dim V_5 = 39{,}657$ on both, profile $(0, 0, 2, 150, 4021, 39657)$, part ranks 19,864 + 19,793; every closure is closed by a deterministic sweep of all its low rows (3727, resp. 4021). Since $\binom{24}{\le 5} - 2^{13} = 47{,}263 > 32{,}057$ and $\binom{26}{\le 5} - 2^{14} = 67{,}298 > 39{,}657$, LFD ≥ 6 at $k = 12$ and $k = 13$ as well.

A unit instance over $p = 65537$. Let $p = 65537 = 2^{16} + 1$, $E : y^2 = x^3 + x$ and $R = (4271, 11720) \in E(\mathbb{F}_p)$. The same (a, b, R) is used at every depth. The point of this instance is that its ideal is the unit ideal at every depth $k \le 15$, by the following lemma.

Lemma 10 (A unit instance at depths up to fifteen; written)**.** *Let $X = 4271$ and let $H \subset \mathbb{F}_p^*$ be the subgroup of order 2^k with $2 \le k \le 15$. There is no pair $(x_1, x_2) \in H^2$ with $S_3(x_1, x_2, X) = 0$. Hence the two-summand chain system of E with target abscissa X has no scalar solution, and its ideal is the unit ideal, at every such depth.*

Proof. One checks $X^{(p-1)/2} = -1$, so X is not a square in $\mathbb{F}_p$, and $X^3 + X = Y^2$ with $Y = 11720$. The group $\mathbb{F}_p^*$ is cyclic of order 2^{16} and its squares form the subgroup of order 2^{15}, so H consists of squares for $k \le 15$. The roots $\pm i$ of $x^2 + 1$ have order 4, hence are squares, so $X \notin \{0, \pm i\}$ and therefore $Y \ne 0$. Let $\alpha : E(\mathbb{F}_p) \to \mathbb{F}_p^*/\mathbb{F}_p^{*2}$ be given by $\alpha(O) = 1$, $\alpha((0,0)) = 1$ and $\alpha((x, y)) = x$ otherwise. It is a group homomorphism: it is the descent map of the 2-isogeny with kernel $\{O, (0,0)\}$ of a curve $y^2 = x(x^2 + ax + b)$, here with $b = 1$ [21, Prop. X.4.9], and its proof is algebraic and holds over every field of characteristic different from two.

Suppose $S_3(x_1, x_2, X) = 0$ with $x_1, x_2 \in H$. By the defining property of S_3 (which also covers $x_1 = x_2$, where S_3 has degree at most one in the last variable), there are points $P_1, P_2 \in E(\overline{\mathbb{F}}_p)$ with $x(P_i) = x_i$ and $x(Q) = X$ for $Q = P_1 + P_2$, after replacing P_2 by $-P_2$ if necessary. Both P_i are affine and not $(0,0)$, since $x_i \in H$. Let φ be the p-power Frobenius. As $x_i \in \mathbb{F}_p$, $y(P_i)^2 \in \mathbb{F}_p$, so $\varphi P_i = \pm P_i$.

(a) $\varphi P_1 = P_1$ and $\varphi P_2 = P_2$. Then $Q \in E(\mathbb{F}_p)$ with $x(Q) = X$, so $Q = \pm R$ and $\alpha(Q) = \alpha(R) = X$. But $\alpha(Q) = \alpha(P_1)\alpha(P_2) = x_1 x_2$ is a square. This contradicts that X is not a square.

(b) $\varphi P_1 = -P_1$ and $\varphi P_2 = -P_2$. Then $\varphi Q = -Q$, so $y(Q)^p = -y(Q)$, and $y(Q)^2 = X^3 + X \neq 0$ gives $(X^3+X)^{(p-1)/2} = y(Q)^{p-1} = -1$. But $X^3 + X = Y^2$ with $Y \in \mathbb{F}_p^*$, so $(X^3+X)^{(p-1)/2} = Y^{p-1} = 1$.

(c) Otherwise, after exchanging the points, $\varphi P_1 = P_1 \neq -P_1$ and $\varphi P_2 = -P_2 \neq P_2$ (a point with $P = -P$ has both signs and falls under (a) or (b) according to the other point). Then $\varphi Q = P_1 - P_2$, and $x(\varphi Q) = \varphi(X) = X = x(Q)$ gives $P_1 - P_2 = \pm(P_1 + P_2)$, so $2P_2 = O$ or $2P_1 = O$, a contradiction.

So the scalar solution set Z is empty. The chain ideal is radical with all points rational, so the full ideal is the vanishing ideal of Z, the unit ideal. □

A direct enumeration of Z at $k = 10, \ldots, 14$ also finds no point. The lemma is not formalized.

A third engine. The runs below use a third implementation of the closure, written separately from the other two and run on a GPU. It computes the reduced row echelon basis of V_D for the fixed column order (weight descending) of the independent engine above, which we call the reference engine. This basis depends only on the subspace, so equal basis hashes of two engines test the whole computation. The engine was validated by equality of both basis hashes with the reference engine on all cases small enough to run on a workstation (eighteen closures at $k = 6, 7$ and $D = 3, 4, 5$ in four execution modes, the degree-5 closures at $k = 9, 10$, ten cases over $p = 65537$ at $k = 6, 7, 8$, interrupted and resumed runs), by a negative control in which the shifts of one row are withheld (every withholding that changed the result was caught by the sweep), and by a self-test on the GPU that compares CPU, GPU and the reference engine on eight cases. Every run below ends with a deterministic full sweep: every shift of every final low row, in both swap parts, reduces to zero against the final basis, with zero repairs. A space that passes the sweep is closed and contains the generator, so it contains V_D, and every row added by the engine is a reduction residual of a shift of an earlier row, so it lies in V_D; rank, profile and membership of 1 are therefore statements about V_D itself. The system generator (S_3 coefficients, chain normal form, enumeration of Z) is shared with the reference engine and cross-checked against one further implementation; it is not re-derived independently. The $k = 14$, $D = 6$ runs used a variant that moves part of the basis to host memory; it passed the same self-test.

In the other direction, the reference engine reproduces both basis hashes of the GPU engine for $V_5(14)$ on the unit instance below (122,438 columns, dimension 47,849) and confirms $V_6(10) = M_{\leq 6}$ and $V_6(11) = M_{\leq 6}$, so the largest degree-5 closure of that instance is computed by two engines with the same canonical basis. The $k = 14$, $D = 6$ closure of the instance was run twice, with different settings, and both runs give the same two basis hashes. An independent checker, a separate program that shares no code with the engines, read the stored basis W of the second run and verified the containing-space certificate directly: the basis is in reduced echelon form, the generator lies in W, each of the 28 shifts of each of the 73,587 low rows reduces to zero against W in both swap parts (4,120,872 checks, no failure), and $1 \notin W$. Its rank, profile and basis hashes equal the engine's. The description of the checker in the repository states which parts of the engine its author had read.

Results to depth fourteen. All values in this paragraph are exact. On the unit instance the degree-5 closure is closed and misses 1 at $k = 10, \ldots, 15$, with $\dim V_5 =$ 18,569, 25,033, 32,057, 39,657, 47,849, 56,649 and profiles

$$(0,0,2,138,3163,18569), \quad (0,0,2,142,3441,25033), \quad (0,0,2,146,3727,32057),$$
$$(0,0,2,150,4021,39657), \quad (0,0,2,154,4323,47849), \quad (0,0,2,158,4633,56649);$$

for $k \leq 13$ these are the dimensions and profiles of the independent engine on the earlier instances. The degree-6 closure is the whole degree-6 space and contains 1 at $k = 10, 11, 12, 13$ ($\dim V_6 = 60{,}460$, 110,056, 190,051, 313,912). With the exclusion of 1 from V_5, the last fall degree of this system is exactly 6 at $k = 10, \ldots, 13$. The upper side at $k = 11, 12, 13$ is new. At $k = 14$ the degree-6 closure is closed and has dimension 448,922 in the 499,178-dimensional space $M_{\leq 6}$ (codimension 50,256), with low profile $(0, 0, 2, 154, 4451, 73587)$, and it does not contain 1. Since 1 lies in the ideal, LFD ≥ 7 at $k = 14$. By Theorem 15, $1 \notin V_6(15)$ as well, and the ideal is the unit ideal at $k = 15$ (Lemma 10), so LFD ≥ 7 at $k = 15$ on this instance too. The codimension of V_D in $M_{\leq D}$ at the first deficient depth $4D - 10$ is 10, 189, 3131 and 50,256 for $D = 3, 4, 5, 6$.

The degree-seven step at depth fourteen. The space $M_{\leq 7}$ at $k = 14$ has 1,683,218 columns, too many for a direct closure on the hardware we used. The exact value was obtained by a degree lift, which computes V_{D+1} from a closed V_D without a closure in $M_{\leq D+1}$; the full statement, proof and algorithm are in the repository (Section 11). Call a weight-D monomial regular if its swap orbit is a pivot column of V_D in both swap parts, and a weight-$(D+1)$ monomial regular if all its weight-D divisors are. Modulo V_D, a shift of a basis row has at most one regular coordinate of weight $D + 1$. The regular top columns are therefore eliminated by subtracting generators that share a column, and what remains is a subspace K of the small space T spanned by the non-regular top columns and the free columns of V_D. The elements of K without top part are the falls. They are added to V_D and shifted in turn, until no new fall appears. At that fixed point $V_{D+1} \cap M_{\leq D}$ is V_D plus the falls, $\dim V_{D+1} = \dim V_D + \sum_q (r_q + \dim K_q)$ over the two swap parts q, with r_q the number of regular top columns of the part, and $1 \in V_{D+1}$ if and only if $1 \in V_D$ or the class of 1 is a fall. The proof is written and not formalized.

On the unit instance the lift started from the checked basis of $V_6(14)$, which has 25,136 + 25,120 free columns. There are 1406 non-regular monomials of weight 6; each part has 586,501 regular and 5519 non-regular columns of weight 7, and $\dim T = 30{,}655$ and 30,639. The lift reached its fixed point in three rounds, with 494 + 494, 12,260 + 12,260 and 12,382 + 12,366 new falls: every free column of V_6 becomes a fall, $K = T$ in both parts, and

$$\dim V_7(14) = 448{,}922 + 2 \cdot 586{,}501 + 30{,}655 + 30{,}639 = 1{,}683{,}218 = |M_{\leq 7}|.$$

So $V_7(14)$ is the whole space and contains 1, and with $1 \notin V_6(14)$ the last fall degree of this system is exactly 7 at $k = 14$ (751 seconds on the GPU). The coverage check of the computation held: every regular weight-7 column was met in exactly one batch, with all seven of its generators.

The lift agrees with the direct closure in 170 runs over 37 cases with $D \leq 5$ and $k \leq 11$, over the primes 12289, 15361, 65537 and 16777213 (for the last, as polynomial systems, since $4 \,\|\, p-1$): the dimension, the profile, the membership of 1 and the canonical hash of $V_{D+1} \cap M_{\leq D}$ in both parts are equal in every run. The cases include instances with solutions, where the lifted space equals the ideal part, and the transitions at which 1 enters, $D = 4 \to 5$ at $k = 6, \ldots, 9$ and $D = 5 \to 6$ at $k = 10$; the last is the analogue of the present step one degree lower, and the lift also reproduced it on the GPU. Negative controls that withhold the iteration of the falls, or all shifts of one index, change the result whenever the withheld rows are not redundant. A further control repeats the step $D = 5 \to 6$ at $k = 10$ on the random curve over $p = 65537$, where the system has 14 solutions and the lifted space can never exceed the ideal: the lift gives $\dim V_6 = 60{,}446 = 60{,}460 - 14$ without 1, with K of codimension 7 in T in each part, and equals the direct closure in dimension, profile and both hashes.

A control at $k = 14$ was then run on the random curve over $p = 114689$ described below, which has 2370 solutions. There the lifted space lies in the ideal, so its dimension is at

most 1,683,218 − 2370 = 1,680,848, and a lift that overshoots would be caught. From a kept, closed $V_6(14)$ of that instance (dimension 448,922, with the profile of the unit instance) the lift reached its fixed point in three rounds, with 494+494, 12,260+12,260 and 11,197+11,181 new falls and K of codimension 1185 in T in each part. It gives $\dim V_7(14) = 1{,}680{,}848$, exactly the bound, without 1, and $\dim(V_7 \cap M_{\leq 6}) = 496{,}808 = 499{,}178 - 2370$. The evaluation of $M_{<6}$, and hence of $M_{<7}$, on Z has rank 2370, computed directly with the degree-6 closure (`experiments/ic-bounds-k14/relaunch/fetched/proper-p114689-k14-D6.json`); the dimension count alone gives only the trivial bound ≤ 2370. So V_7 is the degree-7 part of the ideal, and the local-to-global criterion gives LFD ≤ 7; since V_6 is 47,886 dimensions short of the degree-6 part of the ideal, LFD $= 7$ exactly on this curve as well (4679 seconds; every regular column covered once).

The lift was then run on the second unit instance, $y^2 = x^3 + 3x$ over $p = 163841$ (below). A second run of its degree-6 closure, with the basis kept, reproduced both basis hashes of the first run; the independent checker passed on the kept basis; and the lift reached its fixed point in three rounds with the same counts as over $p = 65537$ ($K_\pm = T_\pm$, new elements $494 + 494$, $12{,}260 + 12{,}260$, $12{,}382 + 12{,}366$): $V_7(14) = M_{\leq 7}$ and $1 \in V_7(14)$, so LFD $= 7$ exactly on this instance as well (677 seconds).

Unlike the closures above, the lift does not end with a sweep in $M_{\leq 7}$ and produces no basis of V_7. The value 7 therefore has no independent certificate: it rests on the written proof, on the assertions of the runs, on the comparisons just listed and on the control, and it is one implementation, run once on each of three instances.

A random-curve replication. The same engine, on $y^2 = x^3 + 4021x + 9917$ over $p = 65537$ (seed 12, target $(1424, 826)$, 1028 solutions at $k = 13$ and 4068 at $k = 14$), gives the same degree-5 dimensions and profiles, 39,657 at $k = 13$ and 47,849 at $k = 14$, against $\dim(I \cap M_{\leq 5}) = 82{,}654$ and 118,370. At $k = 13$ the degree-6 closure has dimension $312{,}884 = 313{,}912 - 1028$, equal to the degree-6 part of the ideal, and the evaluation of $M_{<6}$ on Z has rank 1028, so the local-to-global criterion gives LFD $= 6$ exactly on this instance at $k = 13$. At $k = 14$ the degree-6 closure of this instance is closed and has dimension 448,922, with profile $(0, 0, 2, 154, 4451, 73587, 448922)$ and part ranks 224,688+224,234: the same dimension, profile and part ranks as on the unit instance. The degree-6 part of the ideal has dimension $499{,}178 - 4068 = 495{,}110$ (the evaluation of $M_{\leq 6}$, and already of $M_{<6}$, on Z has rank 4068), so V_6 misses it by 46,188 dimensions and LFD ≥ 7 at $k = 14$ on the random curve as well. As for V_5, the dimension of V_6 at these depths does not depend on the instance.

Two more primes. The engine was also run over $p = 163841 = 5 \cdot 2^{15} + 1$ and $p = 114689 = 7 \cdot 2^{14} + 1$. Over $p = 163841$ the instance is $y^2 = x^3 + 3x$ with target $(73730, 28291)$. Its ideal is the unit ideal for $k \leq 14$: the proof of Lemma 10 applies with $X = 73730$, a non-residue with $X^3 + 3X = 28291^2 \neq 0$, because the subgroup of order 2^k lies in the squares for $k \leq 14$; the enumeration of Z at $k = 13, 14$ also gives no point. At $k = 13$, $\dim V_5 = 39{,}657$ without 1 and $V_6 = M_{\leq 6} \ni 1$, so LFD $= 6$ exactly. At $k = 14$, $\dim V_5 = 47{,}849$ and V_6 is closed of dimension 448,922 without 1, so LFD ≥ 7. Profiles and part ranks equal those over $p = 65537$ at both depths.

Over $p = 114689$ the instance is the random curve $y^2 = x^3 + 31415x + 92653$ with target $(80359, 998)$ (seed 22; 594 solutions at $k = 13$ and 2370 at $k = 14$). At $k = 13$, $\dim V_5 = 39{,}657$, and V_6 is closed of dimension $313{,}318 = 313{,}912 - 594$ with $\dim(V_6 \cap M_{\leq 5}) = 83{,}088 = 83{,}682 - 594$. The evaluation of $M_{<6}$ on Z has rank 594, computed directly (`experiments/ic-bounds-k14/runs/proper-p114689-k13-D6.json`); since V_6 lies in the ideal, V_6 equals the degree-6 part of the ideal; V_5 is 43,431 dimensions short of the degree-5 part. The local-to-global criterion gives LFD $= 6$ exactly, as on the random curve over $p = 65537$. At $k = 14$,

$\dim V_5 = 47{,}849$ with the same profile, and the degree-6 closure is closed of dimension 448,922, with the profile and part ranks of the other instances, against 496,808 for the degree-6 part of the ideal, so LFD ≥ 7; the control lift above gives LFD $= 7$. The degree-6 closures thus agree on four instances over three primes at both depths.

The recursion. Write $p_j(k) = \dim(V_5(k) \cap M_{\leq j})$. On the random curve over $p = 65537$ the degree-5 closure was also computed at $k = 15$, with the profile of the unit instance, and at $k = 16$, with $\dim V_5 = 66{,}073$ and profile $(0, 0, 2, 162, 4951, 66073)$. These profiles satisfy

$$p_j(k+1) = p_j(k) + 2\,p_{j-1}(k) + p_{j-2}(k) \qquad (j \leq 5,\ k = 10, \ldots, 15),$$

with $p_{-1} = p_{-2} = 0$, for instance $18{,}569 + 2{\cdot}3163 + 138 = 25{,}033$, $25{,}033 + 2{\cdot}3441 + 142 = 32{,}057$, $32{,}057 + 2{\cdot}3727 + 146 = 39{,}657$, $39{,}657 + 2{\cdot}4021 + 150 = 47{,}849$, $47{,}849 + 2{\cdot}4323 + 154 = 56{,}649$ and $56{,}649 + 2 \cdot 4633 + 158 = 66{,}073$; for $j \leq 1$ both sides are 0 and for $j = 2$ both are 2. This is the dimension count of the free extension of Theorem 13 by one more level in each block (patterns of weight 0, 1, 2 on the two new variables), and the same recursion holds for $D = 4$ from $k = 6$ to 13. Unrolled, it gives $d_5(10 + n) = 18{,}569 + 2n \cdot 3163 + \binom{2n}{2}138 + 2\binom{2n}{3}$, a cubic in k (third difference 16) against the quintic $\binom{2k}{\leq 5}$; the values at $k = 15$ and 16 had been predicted from this closed form before they were computed. The recursion is an exact observation on computed closures, not a theorem, and it is an identity of dimensions at its first step: the free extension of $V_5(10)$ to depth 11 is not closed (the sweep of the free-extension space leaves nonzero residuals on all five instances tested) and is a different space from $V_5(11)$ of the same dimension. The free extension of $V_5(11)$ to depth 12 is closed, contains the generator and excludes 1 on those five instances, and on the three for which $V_5(12)$ was computed it has the same canonical basis hash as $V_5(12)$. Closedness at depth 12 is the hypothesis of Theorem 13; applied to this computed base, and not to a kernel-checked one, it gives $1 \notin V_5(k)$ and $V_5(k) \subseteq W_k(V_5(11))$, the construction of the theorem with $D = 5$ and base depth 11, for every $k \geq 12$ on those instances, as polynomial systems; over $p = 12289$ they are subgroup systems only for $k \leq 12$. So the free-extension structure of V_5 starts at depth 11 as a base, and the recursion of dimensions starts one depth earlier.

Limits. The lower bound LFD ≥ 7 at $k = 14$ rests on closed degree-6 closures of four instances: two unit instances with $j = 1728$, over $p = 65537$ and $p = 163841$, and two random curves, over $p = 65537$ and $p = 114689$. All were computed by one engine. On the two unit instances the closure is two runs with equal basis hashes and an independent check of the stored basis; on the random curve over $p = 114689$ it is one run with the same independent check; on the random curve over $p = 65537$ it is one run. The upper bound LFD ≤ 7 at $k = 14$ is the lift on three instances (the two unit instances and the random curve over $p = 114689$), one run each, without a certificate, as described above. For $D = 6$ the reference engine confirms the full closure at $k = 10$ and $k = 11$; for $k \geq 12$ there is no second engine. No $k = 14$ degree-6 basis (about 45 GB each) was kept, only the hashes and the checker's record. The lift record names its input basis by a hash over file names, sizes and digests, and the checker record by a manifest hash and per-file digests without sizes, so the two cannot be matched hash for hash; they agree in checkpoint slot, source identifier, rank and free-column counts. Beyond $k = 14$ only degree-5 closures were computed, at $k = 15$ on two instances and at $k = 16$ on one; nothing is known about V_6 and V_7 there beyond the consequences of Theorem 15.

The growth law, as a conjecture. The exact values 4, 5, 6 and the value 7 at $k = 14$ agree with $\mathrm{LFD}(k) = \lfloor (k + 14)/4 \rfloor$ for $2 \leq k \leq 14$. We state this law as a conjecture for generic two-summand prime chain systems, not as a theorem: it was fitted to the exact values at $k \leq 13$, and the exceptional systems of Section 6 already

deviate from it. For unit ideals it reduces, by Theorem 15, to the two finite statements $1 \in V_D(4D-11)$ and $1 \notin V_D(4D-10)$ for every D. Six predictions were frozen in `experiments/ic-bounds-freeext/predictions-2026-11-10.json` before the runs. Four concern depths reported here: $\dim V_5(14) = 47{,}849$ with profile $(0, 0, 2, 154, 4323, 47849)$; $V_6(k)$ equal to the degree-6 part of the ideal for $k \le 13$, with $1 \in V_6(13)$ on unit ideals; $V_6(14)$ a proper subspace not containing 1 on unit ideals; and a table of values of the law, which includes $\mathrm{LFD}(14) = 7$. All four held on the unit instance; the first also held on the random curve, and the same statements hold at the depths computed on the unit instance over $p = 163841$. The value 7 at $k = 14$ was therefore predicted, and was computed afterwards by the lift. Two predictions concern depths not computed: the closed form for $\dim V_5$ at $k = 17, 18$, and the base depth of the free extension of V_6 (`d6-base-15`). That the predictions preceded the runs rests on the dates recorded in the files. What remains untested is every depth beyond 16 for V_5 and beyond 14 for V_6, the value at $k = 14$ by a second method, and uniformity over curves, primes and targets: the degree-6 evidence at depths 13 and 14 is two unit instances with $j = 1728$ and two random curves, over three primes, and the degree-7 evidence is three of these instances.

Three summands. For $m = 3$ the direct presentation $\{3k \text{ chains}, \mathrm{NF}(S_4(x_1, x_2, x_3, t))\}$, with $S_4 = \mathrm{Res}_X(S_3(x_1, x_2, X), S_3(x_3, t, X))$, has exact LFD 7 at $k = 3$ and 4 on all 234 seeded cases (five primes, six curves, up to four targets), and $7 = 3k + 1$ at $k = 2$ in 105 of 116 cases, so the kernel-checked bound $mk + 1$ is attained there (1092 values were recomputed by an independent implementation). The exceptions have $\deg \mathrm{NF}(S_4) = 5$, which happens exactly when the coefficient $-8(5t^3 + 3at + 6b)$ of $x_1^3 x_2^3 x_3^3$ vanishes.

Theorem 11 (Carry lower bound; kernel-checked)**.** *Let G be a reduced-ideal-trivial set with $0 \notin G$, g a reduced polynomial of degree d, and h a reduced polynomial of degree at most d with $h - a\,g \in \langle G \rangle$ for some a and $h \neq c\,g$ for every constant c. Then* $\mathrm{LFD}(G \cup \{g\}) \ge d + 1$. *For three summands, every field with $2 \neq 0$, every depth $k \ge 3$ and $5t^3 + 3at + 6b \neq 0$, the direct system $G_{3,k} \cup \{\mathrm{NF}(S_4)\}$ has* LFD ≥ 7.

Proof. By `FiniteQuotient.closure_le_quotient_span`, V_d lies in $(\langle G \rangle \cap M_{\le d}) + Kg$; if h were there, the reduced element $h - cg$ would lie in $\langle G \rangle$ and vanish (`CarryLower.not_hasLFDBound_of_carry`, `chain_carry_lower`). For $m = 3$ take $g = \mathrm{NF}(S_4)$, of degree exactly 6 because $(3, 3, 3)$ is its only possible weight-6 chain mask, and $h = \mathrm{NF}(u_{1,0}\, g)$, of degree at most 6. For $k \ge 3$ every exponent of S_4 and $x_1 S_4$ is below 2^k, so the chain normal form is injective on these supports and $h = cg$ would force $x_1 S_4 = c S_4$, i.e. $S_4 = 0$. The expansion of S_4 (125 coefficients) and its $(3, 3, 3)$ coefficient are proved by `ring` (`M3CarryLower.m3_lfd_ge_seven`, `prime_field_m3_lfd_ge_seven`). □

The split presentation with a free auxiliary variable has LFD 6 at $k = 4$ and in 111 of 117 cases at $k = 3$ (5 in the other six).

The value does not move with the depth in the range we can reach. On twelve further instances (eight primes with $32 \mid p - 1$, random curves, random and planted generic targets, the same curve and target at every depth) the exact LFD is 7 at every $k = 2, \ldots, 5$: V_6 is spanned by the generator alone, and V_7 is the whole degree-7 part of the ideal, $\dim V_7 = |M_{\le 7}| - |Z|$ with $M_{<7}$ evaluating onto Z ($|Z|$ between 12 and 544 at $k = 5$, where $|M_{\le 7}| = 16{,}384$). So for three summands, at $k \le 5$, the only fall is at $\deg g + 1$, where for two summands it is at $\deg g + 2$ and beyond, and at equal depth the three-summand value is 3 above the two-summand value at $k = 2, \ldots, 5$. No degree from 7 up falls behind the ideal at $k \le 5$, so the free-extension recursion of the two-summand closures had nothing to act on there.

At $k = 6$ the degree-7 closure, computed on one instance, does fall behind the ideal. The instance is $p = 7681$, $y^2 = x^3 + 1876x + 2678$, target abscissa 6223, with $|Z| = 48$ scalar

solutions and $|M_{\le 7}| = 63{,}004$ columns in 18 chain variables. Here $V_6 = \mathrm{span}(g)$ as at smaller depths, but V_7 is closed, by a deterministic full sweep, with $\dim V_7 = 19{,}958$, low profile $(0, 0, 0, 0, 0, 3, 2375)$ and $1 \notin V_7$, against $\dim(I \cap M_{\le 7}) = 62{,}956$. So an element of the ideal of degree at most 7 lies outside V_7, and

$$\mathrm{LFD} \ge 8 \quad \text{at } k = 6$$

for this instance (an exact computation, on one instance and with one engine). This is the first depth at which a three-summand value above 7 is established on any instance, and it is the depth at which the two-summand degree-4 closure first falls behind. The exact value at $k = 6$ would need the degree-8 closure ($|M_{\le 8}| = 106{,}762$ columns), which was not computed, and the only upper bound there is the kernel-checked $3k + 1 = 19$. There is no computation at $k \ge 7$, no replication of the $k = 6$ run on a second instance or engine, and no prediction for the depths beyond 6.

More summands. The two bounds used for three summands apply to every m. Corollary 7 gives $\mathrm{LFD} \le mk + 1$. The first statement of Theorem 11, applied to $g = \mathrm{NF}(S_{m+1})$ and $h = \mathrm{NF}(x_1 g)$, gives $\mathrm{LFD} \ge \deg g + 1$ whenever $\deg h \le \deg g$ and h is not a constant multiple of g. By Lemma 6, $\deg g \le m \min(k, m-1)$, so the two bounds can meet only for $k \le m - 1$, and they do meet when g has the full degree mk, that is, when the product of all mk chain variables occurs in g.

We computed g exactly for $m = 3, \ldots, 8$, that is for the summation polynomials S_4 to S_9, without expanding S_{m+1}. Modulo the chains a polynomial is a function on H^m, so g is the inverse discrete Fourier transform of the values of S_{m+1} on H^m, and these values follow from Semaev's resultant recursion carried out with numbers. As checks, the values vanish at $t = x(P_1 + \cdots + P_m)$ for random rational points and every $m \le 8$, and for $m = 3$ the computation agrees with the generator degree, the number of solutions and the exact LFD of all 344 three-summand systems above with $k \le 3$. The instances are over the primes 97, 193, 257, 7681, 12289 and 65537, on $y^2 = x^3 + x$, $y^2 = x^3 + 1$ and two random curves per prime, with two random rational targets per curve.

m	polynomial	$k = 2$	$k = 3$	$k = 4$
3	S_4	7 (48 of 48)		
4	S_5	9 (48 of 48)	13 (47 of 48)	
5	S_6	11 (48 of 48)	16 (48 of 48)	21 (8 of 8)
6	S_7	13 (48 of 48)	19 (8 of 8)	25 (2 of 2)
7	S_8	15 (47 of 48)	22 (8 of 8)	—
8	S_9	17 (47 of 48)	25 (2 of 2)	—

Table 7: Exact LFD $mk+1$ of the direct m-summand prime chain system at depth $k \le m-1$, with the number of instances on which it is established. Dashes: not computed (more than 2^{24} grid points).

On 409 of the 412 instances with $2 \le k \le m - 1$ the generator has degree mk and the two hypotheses on h hold, so $\mathrm{LFD} = mk + 1$ exactly (Table 7, Figure 4). The dimension bound is therefore attained by summation systems themselves for every m from 3 to 8, with values up to 25, and not only by the abstract Boolean family of Table 1. The attainment is of the simplest kind: at budget $\deg g$ the generator cannot be multiplied, so the lower bound is one more than the input degree, and with $\deg g = mk$ it meets the upper bound $\max(mk + 1, \delta)$. These systems have one fall, directly above the input degree; the agreement says nothing about how the closures grow, and it is not comparable with the two-summand profile, whose falls lie above $\deg g + 1$. The three exceptions are unit ideals on $y^2 = x^3 + x$ with $\deg g = mk - 1$;

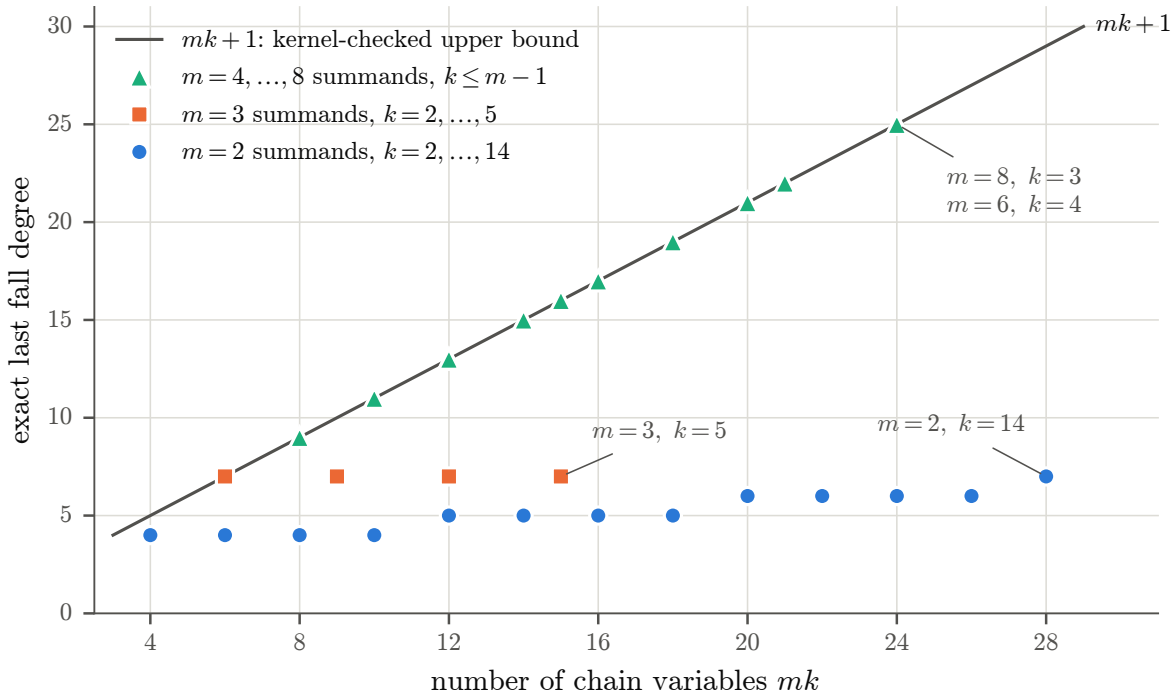


Figure 4: Exact LFD against the number mk of chain variables. With $m \ge 4$ summands at depth $k \le m-1$ (triangles, Table 7) and with three summands at $k=2$, the value lies on the bound $mk+1$. With three summands at $k=3,4,5$ (squares) and with two summands (dots, Table 6) it stays far below.

the argument decides nothing for them. For $k \ge m$ the generator has degree $m(m-1)$ on every instance computed, which gives LFD $\ge m(m-1)+1$: the bound 7 for three summands at $k=3,4$ (96 instances), in agreement with the exact values above, and LFD ≥ 13 for four summands at $k=4,5$ (56 instances), where the upper bounds are 17 and 21. For $m \ge 4$ the lower bounds use the kernel-checked statement with hypotheses established by this exact computation and not by the kernel. The largest cases rest on few instances (sixteen with 2^{20} or 2^{21} grid points, four with 2^{24}), depths with more than 2^{24} grid points were not computed, and no exact value is known for $k \ge m$ when $m \ge 4$.

The chain bridge and the depth reduction. Two kernel-checked statements turn the group-algebra computations of this section into statements about the ordinary closure. Write NF for the graded remainder modulo the chains and $V_D^q(F)$ for the closure in the reduced space under the rule $w \mapsto \mathrm{NF}(x\,w)$ for $\deg w < D$.

Theorem 12 (Chain bridge; kernel-checked)**.** *For every field with $2 \ne 0$, all m, k, every input set $F = G_{m,k} \cup R$ and every $D \ge 2$: $p \in V_D(F)$ if and only if $\deg p \le D$ and $\mathrm{NF}(p) \in V_D^q(F)$ (`ChainClosureBridge.chain_mem_closure_iff`).*

The direct-sum form $V_D(F) = (\langle G_{m,k}\rangle \cap M_{\le D}) \oplus V_D^q(F)$ follows at once; it is written, not a separate Lean declaration.

This proves the premise used throughout the computations, that working modulo the chains gives the ordinary closure for $D \ge 2$, which had been checked only numerically. The second statement is the reduction of all depths to one finite base. For a base depth k_0, a degree D and a subspace S of the depth-k_0 chain algebra in degree at most D, let $W_k(S) = \bigoplus_q \tau_q(S \cap M_{\le D-\mathrm{wt}\,q})$, the sum over high exponent patterns q on levels $k_0,\ldots,k-1$ of the translates of S. Below, $k_0 = 7$ and $D = 4$ unless stated otherwise.

Theorem 13 (Depth reduction; kernel-checked). *Let K be a field with $2 \neq 0$. For every subspace S and all k_0, D, m: if $W_{k_0+1}(S)$ is closed under the admissible variable products at depth $k_0 + 1$, then $W_k(S)$ is closed at every depth $k \geq k_0 + 1$, and $1 \in W_k(S)$ if and only if $1 \in S$ (`DepthReduction.W_closed`, `single_zero_mem_W_iff`). If moreover S contains the generator, then the normal forms of $V_D(k)$ lie in $W_k(S)$ for every $k \geq k_0 + 1$ (`ShiftWitness.normalForm_mem_map_W`). Consequently, for two-summand prime chain systems: if some S at depth 7 contains the generator, excludes 1, and has $W_8(S)$ closed, then $1 \notin V_4$ at every depth $k \geq 8$, and a unit ideal has* LFD ≥ 5 *at every depth $k \geq 8$ (`ChainModelMap.n5_lfd_ge_five`, `prime_field_n5_lfd_ge_five`).*

The hypotheses are three finite facts about one depth-7 space. For two concrete instances the kernel checks them without `native_decide`: $p = 40961$ with $(a, b, t) = (21223, 9887, 4748)$, where $S = V_4(7)$ has dimension 829 in 1471 columns, and a small analogue $p = 11$, $(a, b, t) = (2, 5, 7)$, with the same dimension profile. The certificate packs each echelon row into one large integer, so that a membership test is a few big-integer multiply-and-adds followed by a check of each free slot modulo p. The closedness of $W_8(S)$ reduces, through the component formulas of the reduction, to 3220 such memberships at depth 7, about $2 \cdot 10^7$ kernel operations, which run in about 14 minutes per instance in 42 `decide +kernel` chunks. The result is unconditional for these instances: $1 \notin V_4(k)$ at every depth $k \geq 8$ (`N5CertInstanceP40961.not_one_in_closure`, `N5CertInstanceP11.not_one_in_closure`).

The lower bound itself needs no assumption on the ideal. By definition, LFD $> D$ as soon as one ideal element of degree at most D lies outside V_D, and an element of the low box outside S stays outside $W_k(S)$ at every depth. No single shifted generator $x_1^{e_1} x_2^{e_2} g$ of weight at most 4 is such an element (all 131 candidates with $e_i \leq 125$ lie in $V_4(7)$), but a combination $h\,g$ with $\deg h = (46, 46)$ is: it has weight at most 4, 161 monomials, and a nonzero echelon residual. The kernel checks the product in chunks and the residual, which gives, with no hypothesis left,

$$\text{LFD} \geq 5 \quad \text{at every depth } k \geq 8$$

for $p = 40961$, $(a, b, t) = (21223, 9887, 4748)$ and for $p = 11$, $(2, 5, 7)$, as statements about these polynomial systems at every depth. They are subgroup systems only where $2^k \mid p - 1$: for $p = 40961$ that is $k \leq 13$, beyond which the statement concerns the same system over an extension field; $p = 11$ has no subgroup of order 2^k for $k \geq 2$, and $t = 7$ is not the abscissa of a rational point of $y^2 = x^3 + 2x + 5$, so that instance is a test polynomial system rather than a summation system over $\mathbb{F}_{11}$; the Lean statements are `lfd_ge_five` in the modules `N5CertWitnessInstanceP40961` and `N5CertWitnessInstanceP11`. The same statement for any degree D and base depth k_0, conditional on one base certificate per D and with $k_0 \geq 2$, is `GrowthSchema.growth_schema`; only $D = 4$, $k_0 = 7$ is certified.

Theorem 14 (An exact value above four; kernel-checked). *For $p = 1153$, the curve $y^2 = x^3 + 139x + 1094$ and the target abscissa $t = 185$, the two-summand chain system at depth $k = 6$ has last fall degree exactly 5 (`ExactK6InstanceP1153.exact_lfd_five`).*

This is the first value of the measured profile above 4 that the kernel checks from both sides; the ideal is the unit ideal, which the upper certificate proves and does not assume. The upper half is a derivation of 1 inside V_5 of 83 rows and 39,126 terms, each row a combination of variable shifts of earlier rows within the degree budget; a vector is one big integer with 128×128 slots, so a shift is a multiplication by a power of two, and the wrap $x^{64} = 1$ is folded once per row. The lower half is the containing space $S = V_4(6)$, of dimension 605, with all 1332 shifts of its 111 low rows tested for membership as in the depth-7 certificate. The kernel work is about twelve minutes in 125 `decide +kernel` theorems. The instance was the first unit ideal of a seeded search (small primes with $64 \mid p - 1$ almost never give unit ideals at depth 6, since about $4096/p$ solutions are expected); its closure dimensions $(0, 0, 2, 111, 605)$

and $\dim V_5 = 1586 = |M_{\leq 5}|$ are the generic depth-6 values. It is one instance: the generic value 5 at depth 6 remains a certified computation.

Theorem 15 (Depth restriction; kernel-checked)**.** *For every field, all a, b, t, every $k \geq 2$ and $D \geq 2$: the substitution $u_{i,k} \mapsto 1$ maps the degree-D closure at depth $k+1$ into the degree-D closure at depth k. Hence $1 \in V_D(k+1)$ implies $1 \in V_D(k)$, and an exclusion $1 \notin V_D(k_0)$ holds at every depth $k \geq k_0$ (`DepthRestrictionM2.restrict_closure`, `not_one_of_le`, `lfd_gt_of_le`). As a short written corollary, not a Lean declaration, the last fall degree is nondecreasing in the depth over any range of depths on which the ideal is the unit ideal.*

The substitution sends the chain equations of depth $k+1$ to those of depth k or to zero and fixes the generator, so it maps derivations to derivations without raising degrees. Applied to Theorem 14 it gives, in the kernel, $1 \notin V_4(k)$ at every depth $k \geq 6$ for that instance (`ExactK6Deeper.not_one_in_closure_four_of_ge`). With the certified exclusion $1 \notin V_5(10)$ on the unit instance, Theorem 15 gives $1 \notin V_5(k)$ for every $k \geq 10$ for the same coefficients (a, b, t); the base fact is a certified computation, not a kernel certificate, and beyond $k = 12$ the system over $\mathbb{F}_{12289}$ is no longer a subgroup system ($2^{13} \nmid p-1$), so there the statement concerns the same polynomial system over an extension field. Likewise the exclusion $1 \notin V_6(14)$ on the unit instance over $p = 65537$ gives $1 \notin V_6(k)$ for every $k \geq 14$ for the same coefficients (subgroup systems for $k \leq 16$). For unit ideals the theorem also reduces the growth law $\mathrm{LFD}(k) = \lfloor (k+14)/4 \rfloor$ suggested by the profile to two finite statements per degree, $1 \in V_D(4D-11)$ and $1 \notin V_D(4D-10)$; they hold for $D = 4, 5, 6$ on the computed unit instances ($1 \in V_6(13)$ and $1 \notin V_6(14)$ on the unit instances over $p = 65537$ and $p = 163841$) and are open from $D = 7$ on, where the first test, $1 \in V_7(17)$ against $1 \notin V_7(18)$, is far beyond the present engines; the lift gives $1 \in V_7(14)$ on both unit instances, which $1 \in V_7(17)$ would imply.

What is missing, in order of difficulty.

1. *A genericity theorem.* A proof that for k above a threshold the row reduction of the degree-D closure never pivots on a quantity that vanishes for an actual curve and target outside an explicit exceptional locus, so that $\dim V_D = d_D(k)$ there. The locus must contain the order-two targets and the $(0,0)$-target $j = 1728$ family, where $\dim V_4$ is larger (Section 6). A natural route is to run the closure over the polynomial ring in the coefficients of g and identify the pivots as polynomials in a, b, t; the top coefficients of S_3 are 1, $-2(t^2+a)$ and $-2t$, so already the degree-3 analysis needs $3t^2+a \neq 0$ (Theorem 5). At $k = 6$ and $p = 193$ the exceptional set among the targets computed is known (it was re-verified on a stratified sample of 96 targets including every deviating one): over 1276 targets on 13 curves the exact LFD is 5 except for 12 targets with value 4, which are exactly those with $t = 0$ and $ab = 0$ (3) or with normalized point $(a/t^2, b/t^3)$ in a set C_2 of 190 points of the plane (9); C_2 lies on no curve of degree at most 18, which carries little information, since 190 points in general position lie on no such curve; a cubic C_1 and the tangent line $3t^2+a = 0$ change the closure dimensions but not the LFD. On one C_2-class target at $k = 7$ ($p = 257$) the exact LFD is 5, so the exceptional set depends on the depth at least there, and if C_2 is not the reduction of a curve the theorem must be stated prime by prime. An analysis of this item (with written proofs, not kernel-checked) gives three conclusions. First, the computed closure dimensions lie on both sides of 605 (order-two targets 607, the tangent line 603). Rigorously, only the drop below the computed value on the line $b = t = 0$ is established; if 605 is the generic value, as conjectured, the dimension is semicontinuous in neither direction, and the generic facts cannot be inferred from computed points; what is needed is the generic depth-7 basis over $\mathbb{F}_p(a/t^2, b/t^3)$, whose common denominator has degree at least 480 along a

random line of the plane (an exact lower bound at $k = 6$ and 7, whose records are not in the repository), a certificate of an estimated 10^{10} coefficients that is out of reach of the kernel. Second, full invariance of the closure dimension under $(a, b, t) \mapsto (\mu^2 a, \mu^3 b, \mu t)$ fails for $\mu \notin H$ (730 against 794 on the $t = b = 0$ line at $p = 193$), but the depth-8 certificate is invariant: scaling by μ is the same system with the coset $\mu^{-1}H$ as factor base, and the free extension $W_k(S)$ remains closed in every twisted algebra. Hence one point that passes the depth-7/8 certificate gives LFD ≥ 5 at every depth $k \geq 8$, as polynomial systems (subgroup systems only while $2^k \mid p-1$, so $k \leq 6$ at $p = 193$), on its whole orbit of $p - 1$ scalings (unit ideals; proper ideals when $|Z|$ is below the dimension gap). Third, the strongest exact description of the exceptional set within reach is per prime: the certificate at all 37,249 points of the plane at $p = 193$ is estimated at about 52 CPU-hours.

2. *The polynomials $d_D(k)$.* For $D = 3$ the kernel-checked Theorem 5 proves the consequence that matters, $1 \notin V_3$ at every depth when $3t^2 + a \neq 0$, by showing that no combination of g and its multiples by a variable falls below degree 3. For $D = 4$ the identity $4k^2 + 172k - 571$ is an upper bound per instance for $k \geq 8$, by the free high-level extension of $V_4(7)$, and an equality on the computed range (Section 6); a parametric description of $V_4(7)$, of dimension 829 with a fall of dimension $2k + 86$ above V_3 in the low part, is still missing. For $D = 5$ the closure is full through $k = 9$ and $\dim V_5 =$ 18,569, 25,033, 32,057, 39,657, 47,849, 56,649, 66,073 at $k = 10, \ldots, 16$ satisfy the exact recursion of Section 7, so d_5 is a cubic on the computed range; whether the recursion holds beyond $k = 16$ is untested. For $D = 6$ the closure is full through $k = 13$ and has dimension 448,922 at $k = 14$ on four instances; the deficient codimensions 10, 189, 3131, 50,256 at the first deficient depths $4D - 10$ have no formula.

3. *A lower bound outside the exceptional locus.* An upper bound $\dim V_4 \leq f(k)$ with f quadratic cannot hold for every instance (`PrimeM2Sharp.one_at_four` puts $1 \in V_4$ at every depth on the $(0, 0)$-target family). Proved for all instances outside an explicit exceptional locus, it would give LFD ≥ 5 for every such system with $|M_{\leq 4}| - |Z| > f(k)$, by the exclusion principle of `QuotientLowerBound`. Per instance, Theorem 13 reduces it to three finite facts at depth 7 and 8, and the kernel checks them for concrete instances and proves LFD ≥ 5 there at every depth $k \geq 8$ without further hypotheses (above). What is missing is their genericity in (a, b, t) (item 1), and a base certificate for each degree $D \geq 5$ in the growth schema.

4. *A tight upper bound.* LFD ≤ 5 for all $k < k_0$ needs $d_5(k) \geq \binom{2k}{\leq 5} - |Z|$, i.e. a spanning argument: every weight-≤ 5 polynomial vanishing on Z must be exhibited as a degree-5 derivation. Because the shifts of g alone span only $O(k^3)$ dimensions, the derivation must pass through the fall space $V_5 \cap M_{<5}$ and its shifts; a structural description of that fall space (which carries, from which exponent patterns) is the missing object. The dense final certificates show it exists for $k \leq 9$ but do not describe it, and the depth-ten measurement shows that no such argument can extend to every k: the bound LFD ≤ 5 fails at $k = 10$. The proved upper bounds have slope 1 in k (Theorem 9) and the measured profile has slope about $\frac{1}{4}$; closing that gap needs, for each degree D, the statement $1 \in V_D(4D - 11)$ of Theorem 15 and its analogue for proper ideals, and we found no inductive mechanism relating $V_D(k)$ to $V_{D+1}(k+4)$. The statement $1 \in V_6(13)$ and the exclusion $1 \notin V_6(14)$ are now measured on two unit instances. The exact value 7 at $k = 14$ comes from the degree lift, on both of them and on a random curve; a direct closure in $M_{\leq 7}$ at that depth, which would certify it, is out of reach.

5. *Binary and Koblitz.* The written $\min(k, \ell)+2$ bound is tight in order but not in constant;

the measured two-summand values on the grids are at most 5 through $k = 10$, and the inverse family (a Koblitz curve) has 6 at $k = 10$. On the first grid the 28 systems with value 5, all at $k = 10$, were exactly those with more than 470 solutions. A follow-up shows that this is not a law. Every equation is affine bilinear in the two coordinate blocks and the curve coefficient a does not occur; for a target with $x_R \neq 0$ the trace form $\mathrm{Tr}(\cdot/x_R^2)$ gives one linear equation, which is why a trace-zero coordinate space doubles or empties the solution set. Two candidate laws were fixed before 80 new cases (75 distinct systems) were run: the value as a function of the expected log-count e of solutions alone, and as a function of $k + e$. Both are refuted (11 hits against 36 misses, and 36 against 11, on the 47 systems chosen to separate them): a trace-zero hyperplane at $k = 9$ with about 490 solutions has value 5 in 11 of 11 systems, and unit ideals reach 4 at $k = 7$. There is no bound in the solution count alone, and the subfield $\mathbb{F}_{2^8} \subset \mathbb{F}_{2^{16}}$ with 255 solutions has value 5 at $k = 8$, the inverse-family value $k/2 + 1$. A two-term rule (a floor that grows with k and a term that grows with the solution count) fits the old and new data but was chosen after the check, so it is a conjecture. The pre-certificate closure dimensions depend only on (n, k), the target class and two effective-dimension fields of the record in 1678 of 1775 systems. A tight bound must therefore depend on the coordinate space and on k, not only on the solution count or the dimension. The kernel specialization of the selector theorem is the open formal item of the companion manuscript.

8 Independent checks

The computations were checked in three further ways.

Re-implementation. A separate implementation of the exact closure in the ordinary ring (all monomials of degree $\leq D$, chain or Boolean equations as explicit inputs, variable-wise closure to a fixed point), which shares no code with the closure engines of this paper, recomputed every system within its reach: all 455 prime systems at $k = 2, 3$ and all 985 binary and Koblitz systems with at most 8 Boolean variables. Exact LFD, the list of strict falls, the zero-set size and every per-degree quotient rank agree on all 1440 systems, with no mismatch. A second pipeline that applies the multiplier rule of the Lean definition literally (every monomial multiplier a with $\deg a + \deg f \leq D$) agrees with the variable-wise one, and fourteen further systems at $k = 4, 5, 6$ (two of them with LFD 5 and falls at 4 and 5) also agree.

Proof review. The written arguments (the chain-degree lemma, the soundness of the final verification pass of the shift engine, the local-to-global rank criterion and the depth-ten deduction) were reviewed separately. The equivalence of the quotient-by-chains representation with the ordinary closure for $D \geq 2$, which those arguments use, is kernel-checked (Theorem 12).

Reruns. The depth-ten computation was rerun from scratch on the same instance and reproduces the verified V_5 exactly (same 17 rounds, rank 18,569, same basis hash); the completeness pass of the rerun uses a separate implementation that reduces every shift of every low row against a reduced echelon copy of the basis instead of re-eliminating the stacked matrix. The independent engine of the paragraph on depths eleven to fourteen reproduces all three depth-ten closures and the $\dim V_4$ identity through $k = 13$; it also re-checks the quotient-by-chains premise against the ordinary-ring validator in eleven cases at $k = 4, 5$. The depth-nine systems were recomputed from scratch, and all ten equal their stored records. The GPU engine of the same paragraph reproduces the stored basis hashes of the reference engine on all cases run on both, and it reproduces the dimensions and profiles of the independent engine at $k = 10, \ldots, 13$

for $D = 5$ and the part ranks at $k = 10$ for $D = 6$ (hashes are not comparable across primes). The reference engine in turn reproduces the GPU engine's basis hashes for $V_5(14)$, and an independent checker re-verifies the stored certificate of $V_6(14)$ (Section 7).

9 Related work

The last fall degree was introduced by Huang, Kosters and Yeo [10]; the closure framework used here is that of Huang, Kosters, Yang and Yeo [1], who also show that solving cost depends on the LFD together with the number of solutions [1, Prop. 2.8]; Caminata and Gorla [2] relate LFD to solving degree and the degree of regularity, and their Theorem 2.9 and Example 4.3 give LFD 0 for a Gröbner basis, the case $k = 1$ of the chain system. LFD bounds for Weil-descent systems are due to Huang, Kosters and Yeo [10], Huang [11] and Gorla, Mueller and Petit [12]; Steiner [13] uses the same coprime-leading-term structure to bound solving degrees of iterated systems.

The prime two-power subgroup presentation is due to Petit, Kosters and Messeng [3, §3.3]. Their Table 4 reports Magma's corrected degree of regularity for the same two-summand subgroup-chain presentation, averaged over 100 random instances per depth (the smallest prime p with $2^{n_1} \mid p - 1$ and $p \geq 2^{2n_1}$, a random curve and point, about one solution per instance): 4.00 for $n_1 \leq 5$, 5.00 for $n_1 = 6, \ldots, 9$ and 6.00 for $n_1 = 10, 11$, without an explanation; their Table 5 (isogeny-kernel factor bases) shows 5 up to kernels of order 2^9 and 6 at 2^{10} and 2^{11}. The breakpoints coincide with the exact LFD values certified here, which suggests that LFD, and not a solver artefact, is behind their table; our contribution on this point is the exact certified value (rather than a solver step degree), the exact value 6 on one unit instance at $k = 10, \ldots, 13$ (their $n_1 = 10, 11$ are the first two), the bound LFD ≥ 7 and the lifted value 7 at $k = 14$ on it, and the dimension account of Section 6. Kousidis and Wiemers [14] study first fall degrees of summation polynomials.

Linear-subspace factor bases go back to Diem [19] and Petit and Quisquater [20]; the binary descent with linear coordinate spaces used here is from Faugère, Perret, Petit and Renault [4] after Semaev [5]; symmetrized summation polynomials and small-order torsion are in Faugère, Huot, Joux, Renault and Vitse [6], and automorphism symmetries in Faugère, Gaudry, Huot and Renault [15]. Kosters and Yeo [8] prove that the first fall degree of ordinary binary S_3 systems is 2, show that evaluating summation polynomials is NP-complete, and observe that the degree of regularity grows with n and depends on the choice of coordinate space (their Section 5). Galbraith, Granger, Merz and Petit [16] show that Frobenius-invariant factor bases reduce the number of systems and the linear-algebra cost for subfield and Koblitz curves; our finding that the LFD differences of such bases follow the solution count (Section 6) concerns the degree of a single system and leaves their savings untouched. Galbraith and Gebregiyorgis [7] report Gröbner-basis experiments for binary curves.

Gröbner bases have been formalized in Coq [17], Isabelle/HOL [18] and Lean [22]; we are not aware of a formalization of last fall degree, first fall degree or solving degree. The dimension bound $mk + 1$ is the generic finite-quotient bound and is not new as a bound; what is new is the kernel-checked statement for every input list containing the chains, and its two-summand corollary at every depth.

The Lean development (Lean 4.19.0 with Mathlib) has 513 audited declarations. They depend only on the foundational axioms `propext`, `Classical.choice` and `Quot.sound`, and use no further axioms, no `sorry` and no `native_decide`. The certificates of the sweep are exact arithmetic and can be replayed. The Lean development contains no elliptic curve: S_3 and S_4 enter as explicit polynomials, which are not proved there to be summation polynomials, and the binary and Koblitz bounds are one generic Boolean theorem, the second being the first with $b = 1$. The local-to-global step for prime chains is the same rank argument as the

Boolean one; it is applied by hand and is not an instance of the kernel theorem.

10 Conclusion

For the three presentations of elliptic-curve summation systems studied here, the last fall degree is at most the finite-quotient bound $mk + 1$; for prime chains this is kernel-checked at every depth and for every number of summands. A written proof lowers the two-summand prime bound to $k + 2$ on nonsingular curves. From below, LFD ≥ 4 holds at every depth for two-summand prime chain systems with unit ideal and $3t^2 + a \neq 0$, and LFD ≥ 7 for three summands at every depth $k \geq 3$ when $5t^3 + 3at + 6b \neq 0$; both are kernel-checked.

For two summands the exact values are much smaller than the upper bounds. Over prime fields they are 4, 5, 6 and 7 on the depth ranges 2–5, 6–9, 10–13 and 14, with the exceptions of Section 6; they reproduce the degree-of-regularity breakpoints of Petit, Kosters and Messeng and extend them by three depths. The value at $k = 14$ depends on a degree lift without an independent certificate, and every depth beyond 14 is open. The law $\mathrm{LFD}(k) = \lfloor (k + 14)/4 \rfloor$ fits all of these values and remains a conjecture. With three summands the value is 7 through $k = 5$ and at least 8 on one instance at $k = 6$. With three to eight summands and depth $k \leq m - 1$ the bound $mk + 1$ is attained on the instances computed. For binary and Koblitz curves the value is at most 5 through dimension 10 on the grids computed, while the structured inverse family grows as $\lfloor k/2 \rfloor + 1$, and the differences between random and Frobenius-invariant factor bases follow the number of solutions on our grid, which is an association and not a causal statement.

For two summands a tight upper bound is not proved: the proved bounds grow with slope 1 in k and the measured values with slope about $\frac{1}{4}$. Section 7 lists what closing this gap would need.

11 Code and data

The Lean development, the closure engines, the independent checker, the written proofs and the records behind the numbers in this paper are in the companion repository, except where the text says that records were not kept (the selector-bound test, the genericity denominator, and the $k = 14$ degree-6 bases)

`https://github.com/isogeny-labs/lfd-summation-systems`

Paths are relative to its root.

Material	Location
Lean development, its audit and certificate data	`formal/`, `experiments/ic-lean/`, `ic-bounds-cert/`, `ic-bounds-exact-k6/`
Sweep (Sections 5 and 6)	`experiments/ic-curve-sweep/`
Depths 8, 9 and 10	`experiments/ic-chain-depth8/`, `-depth9/`, `-depth10/`
Independent engine, depths 10 to 13; $\dim V_4$	`experiments/ic-bounds-d5/`, `ic-bounds-exact10/`, `ic-bounds-fall/`
GPU engine, degree lift and checker, depths 10 to 16	`experiments/ic-bounds-k14/`
Free extension and predictions	`experiments/ic-bounds-freeext/`
Three summands	`experiments/ic-bounds-m3/`, `ic-bounds-m3-depth/`
Three to eight summands	`experiments/ic-bounds-msum/`
Exceptional locus at $k = 6$	`experiments/ic-bounds-locus/`
Depth-step table ($\dim V_3$, $\dim V_4$, $k = 3, \ldots, 10$)	`experiments/ic-depth-step/`
Binary and Koblitz follow-up	`experiments/ic-bounds-binary/`, `ic-bounds-binary-law/`
Independent ordinary-ring validator	`experiments/ic-curve-validation/`
Written proofs and their reviews	`docs/proofs/`, `docs/reviews-*.md`

The main computations are reproduced by the following commands.

```
python scripts/verify_lean_fall_degrees.py --lake /path/to/lake
python scripts/run_ic_curve_sweep.py --check-only
python scripts/run_ic_chain_depth8.py --check-only
python scripts/run_ic_chain_depth8.py --check-only --depths 9 --primes 7681 \
  --out experiments/ic-chain-depth9 --workers 1 --product-cap 12000 \
  --shift-max-cells 450000000
python scripts/run_ic_chain_depth10_probe.py --tag rerun
python scripts/run_ic_chain_depth10_probe.py --prime 13313 --seed 11 \
  --want-solutions --tag proper-13313-replay
python scripts/run_ic_chain_depth10_probe.py --prime 15361 --a 4021 \
  --b 9917 --seed 12 --tag random-15361-replay
python scripts/run_ic_chain_depth10_degree6.py --combos 3000
python scripts/run_ic_depth_step.py
python scripts/analyze_ic_base_classes.py
python scripts/ic_bounds_msum.py run --max-log2-points 24
python scripts/plot_ic_lfd_figures.py
python scripts/build_ic_curve_paper.py
```

The first command builds the Lean project and prints the axiom dependencies of every audited declaration. The next three regenerate every case of the sweep and of the depth-8 and depth-9 systems and replay every exact closure. The next three rerun the depth-ten closures (about two hours each) and the one after that the degree-six closure (about five hours, up to ten gigabytes). The last five regenerate the depth-step table, the base-class analysis, the more-summand records, the figures and this document. The runs at depths eleven to fourteen over $p = 65537$ need a GPU (the $k = 14$, $D = 6$ closure used a card with 141 GB and moved part of the basis to host memory); their commands, and those of the degree lift, are in the `README.md` and `LIFT.md` of `experiments/ic-bounds-k14/`. The three-summand, exceptional-locus and binary runs are described in the `README.md` of their directories.

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